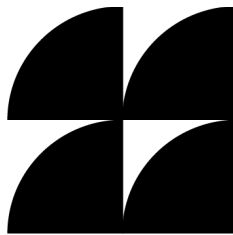


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Contestability Index: Coding Handbook

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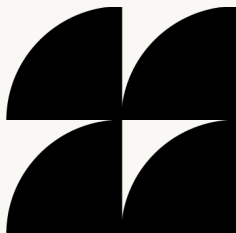


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I. Introduction

This Coding Handbook documents the pilot implementation of the Contestability Measurement Framework and the coding records underlying the measures. Section II sets out the De Jure scoring criteria, data methodology, and legal-form classification. Section III specifies the De Facto scoring rules, indicator structure, market-specific adaptation, gatekeeper selection, and evidence standards. Section IV reports the jurisdiction-level De Jure comparison and presents the factor-level scores, rationales, market coverage, and legal bases for each jurisdiction. Section V applies the framework to the pilot gatekeeper–market observations. For each application, it defines the market scope and applicable mechanisms, reports the aligned market-level De Jure and De Facto assessments, and presents Effective Contestability as a derived summary of the two layers.

[A dedicated project database](#) tracks the full lifecycle of the underlying data. For each item, it records the source (the issuing authority and official publication), the raw collected document, the processed data used for analysis, and a complete version history recording when entries were collected, reviewed, and revised across snapshots. As a result, every entry reported in this Coding Handbook is fully traceable, from the final score back through the processed text to the original official document. The database contents specific to each measure are described in the corresponding section.

II. De Jure Contestability

This section sets out how De Jure Contestability is coded. It defines the scoring criteria applied to every mechanism–jurisdiction cell, describes the data methodology that produces the underlying record, and sets out the legal-form classification used to keep instruments comparable across very different legal systems. The coded results are reported in Section IV.

A. Scoring criteria

Each mechanism–jurisdiction cell rests on three factors, each scored on the same 0 / 0.5 / 1 grid: Existence (whether the relevant obligation is present in the enacted framework), Scope (the range of firms, services, and activities covered), and Enforceability (whether the provision is backed by credible oversight and sanctions).

Factor	Score 0	Score 0.5	Score 1
Existence — is an operative obligation on point present in the enacted framework?	Absent: no binding obligation on point; general competition/consumer law only; drafts; or a designation framework under which no obligation has yet been imposed.	Present but incomplete or formally weak: partial or adjacent coverage; a statutory catalogue still requiring case-by-case activation; administrative guidance of weak formal status; or case-specific precedents/commitments only.	A binding, enacted obligation on point whose content is fixed ex ante by the instrument itself (statute, regulation, court order, or imposed conduct requirement).
Scope — what range of firms, services and activities does the	No coverage: the instrument reaches no relevant firm, service or activity. Scope does not fall	A subset of the provision's domain: firm-specific instruments (remedies, designations, conduct	A rule of general application covering the mechanism's full economic domain within the jurisdiction.

Factor	Score 0	Score 0.5	Score 1
provision reach?	to 0 automatically when Existence = 0. Under legal form F5 an enacted designation framework fixes the addressees without fixing an obligation on point, so Scope may be recorded as 0.5 and Enforceability as 0.5; the final score is 0 through the minimum rule.	requirements), or sectoral/market slices covering only part of the mechanism's economic domain.	
Enforceability — is the provision supported by credible oversight and sanctions?	No enforcement mechanism.	Enforcement requires fresh case-by-case proceedings; guidance without penalties or private enforcement; or machinery untested / not yet effective.	A dedicated regulator (or supervising court) with penalty powers; the obligation is self-executing or directly enforceable, including private enforcement where available.

Final provision score = MIN(Existence, Scope, Enforceability). The minimum is used because the factors do not compensate for one another: an obligation that does not exist is not helped by strong enforcement machinery; a well-drafted obligation binding a single firm is still not a market-wide rule; a broad rule without credible enforcement offers little protection in practice. Scope is judged against the mechanism's own economic domain rather than the whole economy — a marketplace-only statute covers the full domain of a marketplace-native mechanism (Scope = 1 there) but only part of the domain of a mechanism that operates economy-wide (Scope = 0.5).

B. Data methodology

The sixteen mechanisms are grouped under the four dimensions shown in Section IV.A. A mechanism is a stable contestability channel rather than a provision of any single statute; where a channel can be secured through more than one route (for example, alternative distribution through either sideloading or third-party app stores, or technical interoperability through either API access or hardware access), the routes are treated as one mechanism and scored on the strongest available channel.

The data were collected in four steps, so that every score can be traced to a named legal instrument and every instrument to its official text. First, for each jurisdiction the digital-competition instruments in force or pending as of June 2026 (statutes, regulations, court remedies, conduct requirements, administrative rules) were registered with citations and links to official sources, together with a mapping from each mechanism to the articles that govern it. Second, for each mechanism the operative text of every instrument bearing on it was copied verbatim from the primary source, with English translations where needed; scoring works from this primary text, not from secondary summaries. Third, each mechanism–jurisdiction pair was scored on the three factors set out in Section II.A. Fourth, the mechanism scores were averaged into dimension scores and then into the De Jure Contestability Index reported in Section IV.A. The records produced at each step are kept in the project database.

For the De Jure layer, the project database described in the introduction will include: (i) the source register — every statute, regulation, court remedy, conduct requirement, and administrative rule bearing on the sixteen mechanisms, with citation, status, official-source link, and the pending/pipeline instruments tracked alongside those in force; (ii) the original documents — the official texts as published (official gazettes, EUR-

Lex, e-Gov, court dockets), stored in the original language; (iii) the processed versions — the cleaned machine-readable texts and the verbatim provision extracts, with English translations, that scoring works from; (iv) the assessments — all 144 mechanism × jurisdiction records with the three factor scores, the final score, the rationale, the targeted market, and the legal basis, plus the market-coverage matrix; and (v) the history — the review status of each record and its changes across snapshots as instruments are enacted, amended, activated by orders, or repealed (for example, a pipeline bill moving to in force).

C. Legal-form classification

To keep the factor signatures comparable across very different legal systems, instruments are classified into one of eight mutually exclusive legal forms; each form has a typical factor signature from which its score ceiling follows under the MIN rule. This is why, outside the EU and Germany, most non-zero cells sit at 0.5: they rest on firm-specific orders, sectoral statutes, guidance, or case practice rather than on generally applicable ex ante statutes.

Form	Definition and examples	Typical (E, S, Enf)
F1 — General ex ante statute / regulation	Binding obligation of general application, content fixed ex ante, self-executing for all addressees (DMA; Japan's MSCA; Turkey's E-Commerce Law 6563/7416; Korea's TBA Art. 50(1)(9)).	1, 1 (0.5 if sectoral), 1
F2 — Firm-specific binding order	Court remedy, consent decree, settlement, or imposed conduct requirement binding named firms (Epic v. Apple injunction; US v. Google final judgment; CMA conduct requirements).	1, 0.5, 1
F2g — Administrative guidance	Binding in practice but sub-regulatory formal status; no penalties or private enforcement (China's MIIT link-blocking directive; Korea's MSIT pre-installed-app guidelines).	0.5, varies, 0.5
F3 — Ex post enforcement practice	No codified rule on point; obligations arise only from case-specific decisions or commitments under general competition law (JFTC Amazon-parity commitments; KFTC Naver decisions).	0.5, 0.5, 0.5
F4 — Activation catalogue	Enacted statute enumerating prohibited conduct for designated firms, but each prohibition requires a case-by-case activation order (GWB §19a); once issued, the order is reclassified as F2.	0.5, 0.5, 0.5
F5 — Open designation framework	Enacted framework empowering a regulator to impose obligations after designation, where content is not fixed in the statute and no obligation is imposed on point (DMCCA designation without conduct requirements).	0, 0.5, 0.5
F6 — Pending / draft	Bill, draft, or announced regime not yet enacted (Turkish draft Law 4054 amendment; Korean platform bills; Australia's proposed Digital Competition Regime).	0, 0, 0
F7 — None	No instrument on point; general antitrust / consumer law only.	0, 0, 0

Finally, when a market-level index is computed, a sectoral instrument counts at full strength within its own market (for example, Turkey's marketplace rules in the marketplace market), even where the jurisdiction-level table shows a Scope-capped 0.5.

III. De Facto Contestability

This section sets out how De Facto Contestability is coded. It states the scoring criteria, specifies the indicator set, sets out how the set is adapted from one market to the next, states how the gatekeeper and market are selected, and describes the evidence standards. These rules are shared across markets. The market applications are reported in Section V.

A. Scoring criteria

Each dimension is decomposed into De Facto indicators drawn from the Methodology Manual and scored on the 0 / 0.5 / 1 grid used throughout the Index. Each indicator is scored twice, once for the pre-DMA baseline and once for the post-DMA state, so that the change isolates the effect of the regulation from the pre-existing level.

Scores are assigned by the following rule.

Score	Condition
0	The mechanism is absent, prohibited, or obstructed to a degree that leaves no working path.
0.5	A working path exists, subject to friction the gatekeeper imposes or maintains. Fees, warning screens, multi-step procedures, single-presentation prompts, withheld data and unverified self-reporting all fall here.
1	The mechanism operates with no gatekeeper-imposed friction.

Before scoring, each indicator is assigned an applicability status. Applicable means the mechanism operates in the market. Not Applicable means it does not, and the indicator is excluded from aggregation rather than scored zero, since non-applicability carries no information about contestability. Applicable but Missing means the mechanism operates but the evidence required to score it has not been obtained; the indicator is excluded from the denominator and the reason is recorded. Missing status is used where a score would otherwise depend on inference, on general knowledge of a platform, or on the absence of an adverse regulatory finding.

Dimension scores are the unweighted arithmetic mean of the applicable indicators scored. The De Facto Contestability Index is the equal-weighted average of the four dimension scores. The De Jure layer takes the minimum of Existence, Scope and Enforceability because a missing obligation is not cured by enforcement machinery. The De Facto layer averages, because partial progress on several channels adds to observed contestability.

Formal availability does not establish adoption. Where an indicator concerns use or market behaviour, platform documentation alone supports no score above 0.5.

Market-share persistence is excluded as a scored dimension. Following Furman et al. (2019), market share is an outcome of contestability rather than a dimension of it, and including it would introduce circularity into the Index.

B. The De Facto indicator set

The indicator set follows the sixteen observable market conditions in the Methodology Manual. Each is the observable counterpart of one contestability mechanism and is assigned to one dimension. Codes are stable across markets.

Code	Indicator	Mechanism observed
D1 — User Mobility & Switching		
D1.1	User switching behavior	Reduce default bias and enable active user choice
D1.2	Default switching behavior	Ensure users retain control over defaults
D1.3	Application replacement behavior	Allow users to remove pre-installed platform services
D1.4	Data portability usage	Reduce switching costs by allowing users to transfer data across providers
D1.5	Autonomous user decision-making	Prevent manipulative interface designs that impede switching
D2 — Rival Access & Distribution		
D2.1	Alternative distribution adoption	Enable rivals to reach users outside platform-controlled channels
D2.2	Off-platform transactions	Allow business users to communicate and transact directly with customers
D2.3	Commercial pricing flexibility	Prevent contractual restrictions limiting competitive pricing and commercial freedom
D2.4	Rival participation	Ensure rivals can access essential platform services on fair and non-discriminatory terms
D3 — Platform Neutrality & Non-Discrimination		
D3.1	Relative visibility of rival services	Prevent platforms from favouring their own services over rivals
D3.2	Independent service	Prevent platforms from leveraging one service to advantage another

	adoption	
D3.3	Protection of business-user information	Prevent platforms from using non-public business-user data to gain competitive advantage
D4 — Data, Interoperability & Infrastructure Access		
D4.1	Technical interoperability	Enable rivals to interoperate with core platform functionalities and infrastructure
D4.2	Cross-platform messaging connectivity	Reduce network-effect barriers by enabling cross-platform communication
D4.3	Rivals and business-user data utilization	Enable business users to access and use data generated through platform interactions
D4.4	Extent of incumbent data accumulation and combination	Limit structural advantages from exclusive data accumulation or cross-service combination

Each indicator is scored once. Where an observed condition bears on more than one mechanism, it is assigned to the mechanism whose economic function it most directly reflects, and the assignment is documented. The set is not extended with market-specific composites: aggregate measures such as cost of entry, or structural infrastructure costs not imposed by the gatekeeper, are recorded in the evidence column rather than scored.

C. Market-specific variation in the indicator set

The four dimensions and sixteen mechanisms are stable and do not vary. Three things vary across markets.

Applicability. Some mechanisms do not operate in a given market. Cross-platform messaging connectivity applies to number-independent interpersonal communications and to no other market coded here. Parity applies to app stores and intermediation, and reaches search through Shopping, which sits inside the market boundary. Anti-steering reaches any market whose business users are acquired through a Core Platform Service, and so applies in search, though the conduct it observes there differs. Applicability is fixed before scoring and held constant across jurisdictions and periods within a market.

Observation. The same mechanism is observed through different evidence. Default switching is observed in search through choice-screen deployment and the number of steps to change an engine, and in app distribution through the range of reassignable default categories and whether the store is among them.

Competitive weight. The same indicator carries different economic weight across markets. Multi-homing signals little in search, where a user running two engines still sends most queries to one, and signals more in social networking, where accounts on several services coexist and attention divides. Data portability matters more in social networking, where the social graph is the asset that holds users, than in search. Interoperability binds in messaging and operating systems and matters less in marketplaces. Under equal weighting these differences appear in the evidence and in score dispersion, not in the weights. A market-specific weighting scheme requires justification and documentation.

De Facto scores are comparable across jurisdictions and periods within a market, and across gatekeepers within a market where the same indicator set has been applied. They are not comparable across markets. An index of 0.50 in app distribution and 0.50 in search describe different conditions reached through different mechanisms. Cross-market statements should be made at the level of the mechanism.

D. Gatekeeper and market selection

De Facto Contestability is measured for a gatekeeper–market pair. The Methodology Manual requires the gatekeeper to be identified as part of the observation where competitive conditions are platform-specific. An indicator set applies only where the named firm operates the bottleneck the mechanism describes.

A gatekeeper enters a market observation only where it controls a Core Platform Service in that market. Scoring Amazon in online search would be an error of classification: Amazon operates no general search engine, the search indicators would have no object, and the resulting score would measure non-participation rather than low contestability. Alphabet holds the search bottleneck in the European Union. Amazon is the subject in online marketplaces, Meta in social networking and number-independent messaging, and Apple and Alphabet in app distribution and mobile operating systems.

Where a market has more than one gatekeeper, each is coded separately on the same indicator set and reported alongside the other. Averaging them would remove the variation the measure is intended to record. A market-level summary across gatekeepers requires an explicit weighting rule for the relative economic importance of each ecosystem.

Three consequences follow. Applicability is determined against the named gatekeeper's service. Evidence must concern that gatekeeper; a decision addressed to Apple establishes nothing about Alphabet. And a pre-period score of 0 is distinct from non-participation: Apple scores 0 on alternative distribution in December 2023 because it prohibited the channel, which is a finding about contestability, while a firm operating no service in the market is not scored.

E. Evidence standards

Each pair is scored from Commission non-compliance decisions, preliminary findings and specification decisions; gatekeeper compliance reports, developer documentation and published fee schedules; national competition and data-authority actions; judgments of the Court of Justice; and market data from independent providers. Every cell is traceable from its score to the originating document.

A regulatory penalty is not by itself evidence of a market condition. A gatekeeper may be fined because its conduct did not change, in which case the coding is unchanged. A score moves only on documented change in the observable condition. Where a penalty decision also records implemented changes, those records are the evidence, and the decision is cited for the finding that friction persists.

Announced measures that were not operative during the observation window are recorded in the evidence column and do not affect the score.

Three pairs are coded in this version: Alphabet in online search, Apple in iOS app distribution, and Alphabet in Android app distribution. The two app-distribution observations use one indicator set and are reported together so the ecosystems can be compared.

IV. Jurisdiction-Level De Jure Contestability

This section documents the Jurisdiction-Level De Jure Contestability layer of the Index: the existence, scope, and enforceability of the sixteen regulatory mechanisms defined in the Methodology Manual, assessed across the enacted legal frameworks of nine jurisdictions (the European Union, Germany, the United States, Turkey, Japan, Korea, China, the United Kingdom, and Australia) as a post-period snapshot dated June 2026.

This layer applies the scoring criteria, data methodology, and legal-form classification set out in Section II. It is organized as follows. Section IV.A reports the cross-jurisdiction scores by mechanism, together with the dimension averages and the aggregate De Jure Contestability Index; and Section IV.B presents the full scoring table for each jurisdiction, in sequence by country, with the factor scores, rationales, target markets, and legal bases that constitute the evidence for every cell.

A. Cross-jurisdiction comparison

The table reports the final score of every mechanism–jurisdiction cell, organized with the jurisdictions on the horizontal axis and the mechanisms, grouped by dimension, on the vertical axis. The dimension averages and the De Jure Contestability Index — the equal-weighted average of D1–D4 — close the table as its aggregate rows. Jurisdiction codes: EU = European Union, DE = Germany, US = United States, TR = Turkey, JP = Japan, KR = Korea, CN = China, UK = United Kingdom, AU = Australia.

Mechanism	EU	DE	US	TR	JP	KR	CN	UK	AU
D1 — User Mobility & Switching									

Mechanism	EU	DE	US	TR	JP	KR	CN	UK	AU
#1 Active choice	1	1	0	0	1	0	0	0	0
#2 Default-setting restrictions	1	1	0.5	0	1	0	0	0	0
#3 Service removal	1	1	0	0	0.5	0.5	1	0	0
#4 Data portability	1	1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
#5 Choice integrity	1	1	0.5	0.5	0	0.5	0.5	0.5	0
D2 — Rival Access & Distribution									
#6 Alternative distribution	1	1	0.5	0	1	0	0	0	0
#7 Anti-steering	1	1	0.5	0.5	1	0.5	0	0	0
#8 Commercial independence	1	1	0	1	0.5	0.5	0.5	0	0
#9 Platform access	1	1	0	0	0.5	0.5	0.5	0	0
D3 — Platform Neutrality & Non-Discrimination									
#10 Ranking neutrality	1	1	0	1	1	0.5	0.5	0.5	0
#11 Service neutrality	1	1	0	0.5	1	1	0.5	0	0
#12 Business user protection	1	1	0	1	1	0	0.5	0.5	0
D4 — Data, Interoperability & Infrastructure Access									
#13 Technical interoperability	1	1	0	0	1	0	0	0.5	0
#14 Network interoperability	1	1	0	0	0	0	0.5	0	0
#15 Data accessibility	1	1	0.5	1	0.5	0	0	0	0
#16 Cross-service data combination	1	1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
D1 average	1.00	1.00	0.30	0.20	0.60	0.30	0.40	0.20	0.10
D2 average	1.00	1.00	0.25	0.38	0.75	0.38	0.25	0.00	0.00
D3 average	1.00	1.00	0.00	0.83	1.00	0.50	0.50	0.33	0.00
D4 average	1.00	1.00	0.25	0.38	0.50	0.13	0.25	0.25	0.13
DE JURE INDEX (equal-weighted D1–D4)	1.00	1.00	0.20	0.45	0.71	0.33	0.35	0.20	0.06

The same 0.5 can mean different things — a strong law with narrow reach (Turkey’s anti-steering rule, marketplaces only) or broad reach in weak legal form (China’s ranking-neutrality rules) — so the factor scores matter, not just the final score. The per-jurisdiction tables in Section IV.B record the three factors separately, with the rationale, target market, and legal basis for every cell.

B. Jurisdiction scoring tables (scores and evidence)

The tables below present the full scoring record for each jurisdiction, in sequence by country, in the order EU, Germany, United States, Turkey, Japan, Korea, China, United Kingdom, Australia. A note under each country heading states the jurisdiction-wide enforcement machinery. The columns give the mechanism; the final score; the factor scores E = Existence, S = Scope, Enf = Enforceability; the rationale, which records only what deviates from the standard reading of the scores set out in Section II.A (‘—’ indicates that there is no deviation to explain, and for rows scored 0 the absent or pipeline instrument is named in

the Legal basis column); the target market or field; and the full name of the legal basis. Dimension averages and the jurisdiction's De Jure Contestability Index close each table.

1. EU

Enforcement: European Commission (exclusive competence); fines up to 10% of worldwide turnover, 20% for repeat infringements; DMA obligations self-executing on designation.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	1	1	1	1	—	OS, browsers, search (designated CPS)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3)
#2 Default-setting restrictions	1	1	1	1	—	OS, browsers, search, virtual assistants	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3)
#3 Service removal	1	1	1	1	—	OS (designated)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3)
#4 Data portability	1	1	1	1	—	All markets (GDPR) + designated CPS (continuous real-time)	General Data Protection Regulation (Regulation (EU) 2016/679), Art. 20; Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(9)
#5 Choice integrity	1	1	1	1	—	All online platforms	Digital Services Act (Regulation (EU) 2022/2065), Art. 25; Digital Markets Act (Regulation (EU) 2022/1925), Art. 13
D2 — Rival Access & Distribution							
#6 Alternative distribution	1	1	1	1	Merged — governing channel: Sideloaded rights; other component(s): Alternative app store access (final 1.0)	Mobile OS / app distribution; Mobile OS / app stores	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(4)
#7 Anti-steering	1	1	1	1	—	App stores + all designated CPS	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(4)
#8 Commercial independence	1	1	1	1	—	Designated CPS	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(3)
#9 Platform access	1	1	1	1	—	App stores, search, social	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(12)
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	1	1	1	1	Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 1.0)	Search, marketplaces, app stores (designated); Designated CPS + all intermediation (P2B transparency)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(5) Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(5); Platform-to-Business Regulation (Regulation (EU) 2019/1150)
#11 Service neutrality	1	1	1	1	Merged — governing channel: Anti-tying provisions; other component(s):	Designated CPS	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(12)

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
					Anti-bundling provisions (final 1.0)		2022/1925), Art. 5(7) Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(8)
#12 Business user protection	1	1	1	1	—	Designated CPS	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(2)
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	1	1	1	1	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 1.0)	OS, virtual assistants (designated); OS/connected devices	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(7)
#14 Network interoperability	1	1	1	1	—	Messaging (WhatsApp, Messenger)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 7
#15 Data accessibility	1	1	1	1	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 1.0)	Designated CPS; Search (Google)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(10) Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(11)
#16 Cross-service data combination	1	1	1	1	—	Designated CPS + all markets (GDPR)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(2); General Data Protection Regulation (Regulation (EU) 2016/679), Arts. 6-7
D1 average (Final Score)	1.00						
D2 average (Final Score)	1.00						
D3 average (Final Score)	1.00						
D4 average (Final Score)	1.00						
DE JURE INDEX (equal-weighted average of D1–D4)	1.00						

2. Germany

Carries the EU scores: DMA fully applicable; enforcement by the Commission plus the Federal Cartel Office's decentralised powers (11th GWB amendment) and active private enforcement (LG Mainz 2025).

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#1 Active choice	1	1	1	1	—	OS, browsers, search	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3); Act against Restraints of Competition (GWB), §19a(2) available
#2 Default-setting restrictions	1	1	1	1	—	OS, browsers, search	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3); Act against Restraints of Competition (GWB), §19a(2)2
#3 Service removal	1	1	1	1	—	OS	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(3); Act against Restraints of Competition (GWB), §19a(2)3
#4 Data portability	1	1	1	1	—	All markets	General Data Protection Regulation (Regulation (EU) 2016/679); Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(9); Act against Restraints of Competition (GWB), §19a(2)5
#5 Choice integrity	1	1	1	1	—	All online platforms	Digital Services Act (Regulation (EU) 2022/2065), Art. 25; Act against Unfair Competition (UWG)
D2 — Rival Access & Distribution							
#6 Alternative distribution	1	1	1	1	Merged — governing channel: Sideloading rights; other component(s): Alternative app store access (final 1.0)	Mobile OS; Mobile OS / app stores	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(4)
#7 Anti-steering	1	1	1	1	—	App stores+	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(4)
#8 Commercial independence	1	1	1	1	—	All intermediation	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(3) + case law (HRS/Booking line)
#9 Platform access	1	1	1	1	—	App stores, search, social + intermediaries	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(12); Act against Restraints of Competition (GWB), §19a(2)+§20
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	1	1	1	1	Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 1.0)	All markets of designated firms; Same as EU	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(5); Act against Restraints of Competition (GWB), §19a(2)1 Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(5); Platform-to-Business Regulation (Regulation (EU) 2019/1150)
#11 Service neutrality	1	1	1	1	Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 1.0)	All markets of designated firms; Designated firms	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(7); Act against Restraints of Competition (GWB), §19a(2)3 Digital Markets

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
							Act (Regulation (EU) 2022/1925), Art. 5(8); §19a(2)3; private enforcement (LG Mainz Aug 2025 Gmail/Android setup)
#12 Business user protection	1	1	1	1	—	Designated firms	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(2); Act against Restraints of Competition (GWB), §19a(2)4
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	1	1	1	1	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 1.0)	Designated firms; Same	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(7); §19a(2)5 Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(7)
#14 Network interoperability	1	1	1	1	—	Messaging	Digital Markets Act (Regulation (EU) 2022/1925), Art. 7
#15 Data accessibility	1	1	1	1	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 1.0)	Designated firms; Search	Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(10); §19a(2)6/7 Digital Markets Act (Regulation (EU) 2022/1925), Art. 6(11)
#16 Cross-service data combination	1	1	1	1	—	Designated CPS (DMA) + all markets of designated firms (§19a)	Digital Markets Act (Regulation (EU) 2022/1925), Art. 5(2); Act against Restraints of Competition (GWB), Section 19a(2) No. 4
D1 average (Final Score)	1.00						
D2 average (Final Score)	1.00						
D3 average (Final Score)	1.00						
D4 average (Final Score)	1.00						
DE JURE INDEX (equal-weighted average of D1–D4)	1.00						

3. United States

No ex ante statute: obligations are firm-specific court remedies enforced by the supervising courts (contempt powers; the US v. Google final judgment adds a Technical Committee with source-code access). Rows scored 0 have no instrument on point beyond general antitrust law.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0	0	—	—	—
#2 Default-setting restrictions	0.5	0.5	0.5	1	E=0.5: remedy bans exclusive default agreements (contract-side) but creates no user-facing default-change right — adjacent to the provision; S=0.5: Google only; Enf=1: court supervision + Technical Committee (appeal filed Jan 2026 does not stay judgment)	Search (Google only)	US v. Google remedies (final judgment effective 3 Feb 2026; stay denied; appeal pending D.C. Cir.)
#3 Service removal	0	0	0	0	—	—	—
#4 Data portability	0.5	0.5	0.5	1	E=0.5: one-time access/portable copy only — no continuous or API-based portability; S=0.5: California + ~15 states, no federal right; Enf=1: state AG/CPPA enforcement active	All markets, CA + ~15 states	California Consumer Privacy Act, as amended by the California Privacy Rights Act (Cal. Civ. Code §1798.100 et seq.), plus other state privacy statutes
#5 Choice integrity	0.5	0.5	0.5	1	E=0.5: no federal statutory dark-pattern ban — FTC Act §5 unfairness doctrine plus CPRA consent rule; S=0.5: state patchwork, doctrine-dependent federally; Enf=1: active FTC enforcement (e.g. Amazon Prime)	All markets	Federal Trade Commission Act, §5; California Privacy Rights Act (consent obtained through dark patterns is not valid consent); state unfair-and-deceptive-acts-and-practices statutes
D2 — Rival Access & Distribution							
#6 Alternative distribution	0.5	1	0.5	1	E=1: operative instrument is the in-force Oct 2024 injunction (third-party stores, catalog access); the settlement would replace it but is not yet approved (summer 2026 hearing); S=0.5: Google/Android only; Enf=1: court contempt powers	Android only	Epic v. Google permanent injunction (Oct 2024, affirmed 9th Cir. Jul 2025, in force); amended global settlement (filed 4 Mar 2026) pending court approval
#7 Anti-steering	0.5	1	0.5	1	E=1: link-out rights without fees or scare screens; Google side rests on the in-force injunction, not the pending settlement; S=0.5: case-based, not a market rule; Enf=1: court contempt powers	App stores (both duopolists)	Epic v. Apple UCL injunction (contempt order Apr 2025); Epic v. Google injunction (in force; settlement pending approval)
#8 Commercial independence	0	0	0	0	—	—	Rule-of-reason case law only
#9 Platform access	0	0	0	0	—	—	—
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	0	0	0	0	Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.0)	—	—
#11 Service neutrality	0	0	0	0	Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.0)	—	General Sherman Act §1 only —
#12 Business user protection	0	0	0	0	—	—	—
D4 — Data, Interoperability & Infrastructure Access							

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#13 Technical interoperability	0	0	0	0	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.0)	—	—
#14 Network interoperability	0	0	0	0	—	—	—
#15 Data accessibility	0.5	1	0.5	1	E=1: mandated sharing of search index metadata and user-interaction data with court-approved Qualified Competitors, plus syndication; S=0.5: Google only, recipients court-approved; Enf=1: Technical Committee with source-code access + court supervision [Merged — governing channel: Search query and click-data access obligations; other component(s): Business-user data access rights (final 0.0)]	—; Search (Google only)	—; US v. Google remedies (final judgment effective 3 Feb 2026; stay of data-sharing provisions denied; appeal pending)
#16 Cross-service data combination	0.5	0.5	0.5	1	No federal instrument restricts cross-service combination. State privacy statutes impose purpose limitation - collection, use, retention and sharing must be reasonably necessary and proportionate to the disclosed purpose and not further processed incompatibly - which constrains combination without prohibiting it, so Existence is 0.5. Scope is 0.5: a state-by-state patchwork rather than a rule of general application, consistent with the treatment of data portability. Enforceability is 1: the California Privacy Protection Agency and State Attorneys General hold penalty powers.	All markets, CA + ~15 states	California Consumer Privacy Act as amended by the California Privacy Rights Act, Civil Code s.1798.100(c) and s.1798.121; comparable state privacy statutes
D1 average (Final Score)	0.30						
D2 average (Final Score)	0.25						
D3 average (Final Score)	0.00						
D4 average (Final Score)	0.25						
DE JURE INDEX (equal-weighted average of D1–D4)	0.20						

4. Turkey

Enforcement: Ministry of Trade administrative fines plus the ETHS licensing regime (E-Commerce Law 6563 as amended by Law 7416). Rows scored 0 await the draft amendment to Competition Law 4054 (pipeline).

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0	0	—	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
							enacted)
#2 Default-setting restrictions	0	0	0	0	—	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted)
#3 Service removal	0	0	0	0	—	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted)
#4 Data portability	0.5	1	0.5	1	E=1: statutory seller-data access and portability; S=0.5: marketplace sellers only — the portability domain is economy-wide and KVKK grants no general right	Marketplaces only	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022) (seller data portability)
#5 Choice integrity	0.5	0.5	1	1	E=0.5: dark patterns reached by classification as unfair commercial practice, not a dedicated interface-design rule; S=1: all consumer-facing markets; Enf=1: active Advertisement Board enforcement 2024–25	All consumer-facing markets	Regulation on Commercial Advertising and Unfair Commercial Practices (dark commercial patterns added 1 Mar 2022)
D2 — Rival Access & Distribution							
#6 Alternative distribution	0	0	0	0	Merged — governing channel: Sideloaded rights; other component(s): Alternative app store access (final 0.0)	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted)
#7 Anti-steering	0.5	1	0.5	1	E=1: statutory ban on restricting sellers' sales/advertising on other channels; S=0.5: marketplace sellers only — the steering domain centrally includes app stores	Marketplaces only	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022)
#8 Commercial independence	1	1	1	1	E=1: statutory ban on restricting sellers' prices/terms on other channels (wide-MFN ban); S=1: marketplaces are the paradigm domain of parity clauses	Marketplaces only	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022)
#9 Platform access	0	0	0	0	—	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted)
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	1	1	1	1	E=1: structural ban — large ETHS may not sell own-brand goods on their own marketplaces; S=1: marketplaces are a paradigm self-preferencing domain, extended by TCA Google search remedies [Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.0)]	Marketplaces only; —	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022) Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted) (ranking transparency)
#11 Service neutrality	0.5	0.5	0.5	1	E=0.5: limits on conditioning marketplace access on ancillary purchases — narrower than an anti-tying prohibition; S=0.5: marketplaces only — the tying domain spans ID/browser/payment services across CPS [Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.5)]	Marketplaces	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022) E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022) (ad-budget caps; cross-service restrictions)

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#12 Business user protection	1	1	1	1	E=1: explicit ban on ETHS using seller data to compete against sellers; S=1: marketplaces are the paradigm domain of business-user data misuse	Marketplaces only	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022)
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	0	0	0	0	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.0)	—	Draft amendment to the Law on the Protection of Competition No. 4054 (pending, not enacted) —
#14 Network interoperability	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	—
#15 Data accessibility	1	1	1	1	E=1: sellers statutorily entitled to their transaction data including free portability; S=1: marketplaces are the paradigm domain [Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)]	Marketplaces only; —	E-Commerce Law No. 6563, Supplementary Article 2 (added by Law No. 7416, 2022) —
#16 Cross-service data combination	0.5	0.5	1	1	Law 6563 imposes no restriction on combining data across services; its Ek Madde 2 provisions govern use of business-user data in competition and cross-promotion between the intermediary's own environments. The general data protection statute supplies purpose limitation and a consent requirement, which constrains cross-service combination without a platform-specific prohibition, so Existence is 0.5. Scope 1 as a rule of general application; Enforceability 1, the KVKK Board holding penalty powers.	All markets (KVKK)	Law No. 6698 on the Protection of Personal Data (KVKK), Arts. 4 and 5
D1 average (Final Score)	0.20						
D2 average (Final Score)	0.38						
D3 average (Final Score)	0.83						
D4 average (Final Score)	0.38						
DE JURE INDEX (equal-weighted average of D1–D4)	0.45						

5. Japan

Enforcement: JFTC; surcharges up to 20% of relevant Japanese turnover (30% for repeat violations); MSCA in full force since 18 December 2025.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#1 Active choice	1	1	1	1	—	Mobile: browsers, search	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 12
#2 Default-setting restrictions	1	1	1	1	—	Mobile: OS, browsers, search	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 12
#3 Service removal	0.5	0.5	1	1	E=0.5: MSCA Art. 12 gives a default-change/choice framework but no standalone uninstallation article (cf. DMA Art. 6(3)); S=1: covers the provision's domain (mobile OS) for both designated providers	Mobile OS	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 12 default-change framework (no standalone uninstall article)
#4 Data portability	0.5	1	0.5	1	E=1: explicit MSCA data-portability mandate; S=0.5: designated mobile software only — APPI grants no general portability right	Mobile markets only	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), data-portability provisions
#5 Choice integrity	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	—
D2 — Rival Access & Distribution							
#6 Alternative distribution	1	1	1	1	Merged — governing channel: Alternative app store access; other component(s): Sideload rights (final 0.5)	Mobile OS; Mobile OS / app stores	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 6
#7 Anti-steering	1	1	1	1	—	Mobile app stores	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 8
#8 Commercial independence	0.5	0.5	0.5	0.5	E=0.5: no codified rule — obligation exists only through case-specific decisions/commitments; S=0.5: binds only the firms subject to those cases; Enf=0.5: requires fresh case-by-case proceedings	Marketplaces, travel	Antimonopoly Act (Act No. 54 of 1947) enforcement (Amazon Japan parity commitments 2020); JFTC scrutiny
#9 Platform access	0.5	0.5	0.5	1	E=0.5: fair/equitable-treatment duties, narrower than a FRAND access right; S=0.5: mobile app stores only — the FRAND domain spans app stores, search and social	Mobile app stores	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 6 (fair treatment); Act on Improving Transparency and Fairness of Digital Platforms (Act No. 38 of 2020)
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	1	1	1	1	E=1: MSCA Art. 9 ban on preferencing own services in search results; S=1: covers the designated (mobile) search domain — Google designated [Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.5)]	Mobile search (Google); Mobile search; app stores/marketplaces (transparency)	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 9 Smartphone Software Competition Promotion Act (Act No. 58 of 2024), search-display provisions; Act on Improving Transparency and Fairness of Digital Platforms (Act No. 38 of 2020), ranking-transparency provisions
#11 Service neutrality	1	1	1	1	Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.5)	Mobile app stores; Mobile markets	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 8 Smartphone Software Competition Promotion Act (Act No. 58 of 2024)
#12 Business user	1	1	1	1	E=1: MSCA Art. 5 ban on using developers' app data to compete; S=1:	Mobile app stores	Smartphone Software Competition Promotion

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
protection					app-store domain, both designated providers		Act (Act No. 58 of 2024), Art. 5
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	1	1	1	1	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.5)	Mobile OS	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), Art. 7(ii) Smartphone Software Competition Promotion Act (Act No. 58 of 2024)
#14 Network interoperability	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	—
#15 Data accessibility	0.5	0.5	0.5	1	E=0.5: data provisions are partial and transparency-oriented; S=0.5: mobile app stores + marketplace transparency only [Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)]	Mobile app stores; marketplaces (transparency); —	Smartphone Software Competition Promotion Act (Act No. 58 of 2024), data provisions; Act on Improving Transparency and Fairness of Digital Platforms (Act No. 38 of 2020), transparency provisions —
#16 Cross-service data combination	0.5	0.5	1	1	MSCA Art. 5 prohibits designated providers from using data acquired through their specified software to compete with third-party providers, and the JFTC Guidelines extend it to data that has been combined with other data - but the object is business-user data, which is scored at mechanism #12. Nothing in the MSCA restricts the designated provider from combining its own cross-service data. APPI Art. 18 supplies purpose limitation with consent required to exceed the specified purpose, so Existence is 0.5. Scope 1; Enforceability 1 through the Personal Information Protection Commission.	All markets (APPI) + designated mobile software (MSCA)	Act on the Protection of Personal Information (APPI), Art. 18; Mobile Software Competition Act, Art. 5
D1 average (Final Score)	0.60						
D2 average (Final Score)	0.75						
D3 average (Final Score)	1.00						
D4 average (Final Score)	0.50						
DE JURE INDEX (equal-weighted average of D1–D4)	0.71						

6. Korea

Enforcement: KCC for the TBA app-market rules; KFTC case practice under the MRFTA; PIPA portability in force since March 2025 (sectoral). Rows scored 0 have no instrument on point.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0	0	—	—	—
#2 Default-setting restrictions	0	0	0	0	—	—	—
#3 Service removal	0.5	0.5	1	0.5	E=0.5: ministerial guideline/industry agreement — binding in practice, weak formal status; S=1: all carriers/OEMs/Google in the mobile-OS domain; Enf=0.5: no statutory penalties or private enforcement	Mobile OS	MSIT Guidelines on the Installation and Operation of Smartphone Pre-installed Apps (April 2014; industry agreement with carriers/OEMs/Google)
#4 Data portability	0.5	1	0.5	1	E=1: statutory transmission right in force 13 Mar 2025; S=0.5: initial sectors only (healthcare, telecom, energy/finance MyData) — all-sector decree still pending; Enf=1: PIPC enforcement	Healthcare, telecom, energy/finance (MyData) sectors only (initial rollout)	Personal Information Protection Act, Art. 35-2 (in force 13 Mar 2025)
#5 Choice integrity	0.5	1	0.5	1	E=1: statutory dark-pattern provisions (eff. Feb 2025); S=0.5: e-commerce/marketplace scope only — the provision's domain is all consumer-facing services; Enf=1: KFTC enforcement	E-commerce/marketplaces	Act on the Consumer Protection in Electronic Commerce, dark-pattern amendment (eff. February 2025)
D2 — Rival Access & Distribution							
#6 Alternative distribution	0	0	0	0	Merged — governing channel: Sideloading rights; other component(s): Alternative app store access (final 0.0)	—	—
#7 Anti-steering	0.5	0.5	1	0.5	E=0.5: steering protection only implied by the payment-coercion ban — no express steering right; S=1: app stores (the provision's paradigm domain), both duopolists; Enf=0.5: KCC enforcement of outlink restrictions contested	App stores	Telecommunications Business Act, Art. 50(1) (9) (2021 amendment); KCC enforcement
#8 Commercial independence	0.5	0.5	0.5	0.5	E=0.5: no codified rule — obligation exists only through case-specific decisions/commitments; S=0.5: binds only the firms subject to those cases; Enf=0.5: requires fresh case-by-case proceedings	Marketplaces	Monopoly Regulation and Fair Trade Act — KFTC enforcement; platform guidelines
#9 Platform access	0.5	0.5	0.5	1	E=0.5: limited fair-terms duties for app markets, not a FRAND access right; S=0.5: app stores only; Enf=1: KCC/KFTC statutory penalty powers	App stores	Telecommunications Business Act, app-market provisions (Arts. 22-9, 50(1)(9))
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	0.5	0.5	0.5	0.5	E=0.5: no codified rule — obligation exists only through case-specific decisions/commitments; S=0.5: binds only the firms subject to those cases; Enf=0.5: requires fresh case-by-case proceedings [Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.5)]	Search, marketplaces	KFTC Naver decisions (2020); review guidelines KFTC platform review guidelines; Naver precedent
#11 Service neutrality	1	1	1	1	E=1: world-first statutory ban on forcing a specific in-app payment method; S=1: app stores are the paradigm domain of payment tying — both duopolists bound; Enf=1: KCC/KFTC statutory penalty powers [Merged — governing channel: Anti-tying provisions; other component(s):	App stores only; —	Telecommunications Business Act, Art. 50(1) (9) (2021 amendment) —

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
					Anti-bundling provisions (final 0.0)]		
#12 Business user protection	0	0	0	0	—	—	—
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	0	0	0	0	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.0)	—	—
#14 Network interoperability	0	0	0	0	—	—	—
#15 Data accessibility	0	0	0	0	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)	—	—
#16 Cross-service data combination	0.5	0.5	1	1	No competition instrument restricts cross-service data combination. PIPA Art. 18 bars use or provision of personal information beyond the scope of the purpose for which it was collected, absent separate consent, which constrains combination without a platform-specific prohibition, so Existence is 0.5. Scope 1; Enforceability 1 through the Personal Information Protection Commission, which holds penalty powers.	All markets (PIPA)	Personal Information Protection Act (PIPA), Art. 18
D1 average (Final Score)	0.30						
D2 average (Final Score)	0.38						
D3 average (Final Score)	0.50						
D4 average (Final Score)	0.13						
DE JURE INDEX (equal-weighted average of D1–D4)	0.33						

7. China

Hybrid regime: statutes (AML, AUCL, E-Commerce Law) state high-level duties; ministerial rules (SAMR, CAC, MIIT) supply the operative detail; enforcement is discretionary and campaign-cyclical. Rows scored 0 have no instrument on point.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0	0	—	—	—

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#2 Default-setting restrictions	0	0	0	0	—	—	—
#3 Service removal	1	1	1	1	E=1: binding ministerial regulation — non-basic pre-installed apps must be uninstallable; S=1: general application across the mobile-OS domain; Enf=1: binding ministerial regulation; MIIT supervision	Mobile OS	MIIT Interim Provisions on the Pre-installation and Distribution of Mobile Smart Terminal Applications (2016, eff. 1 July 2017)
#4 Data portability	0.5	0.5	1	0.5	E=0.5: transfer right conditional on CAC-specified conditions; S=1: general application (PIPL); Enf=0.5: implementation narrow, administrative discretion high	All markets	Personal Information Protection Law of the PRC, Art. 45(3)
#5 Choice integrity	0.5	0.5	1	0.5	E=0.5: partial provisions scattered across AUCL/CAC/SAMR instruments — no unified dark-pattern ban; S=1: platforms generally; Enf=0.5: administrative enforcement, discretionary	Platforms	Anti-Unfair Competition Law of the PRC (2025 revision, eff. 15 October 2025) Art. 13; Personal Information Protection Law of the PRC; CAC Provisions on the Administration of Algorithmic Recommendations for Internet Information Services (eff. 1 March 2022) Arts. 8/14 (personalization opt-out); SAMR Interim Provisions on Anti-Unfair Competition on the Internet (eff. 1 September 2024) 2024 Art. 18
D2 — Rival Access & Distribution							
#6 Alternative distribution	0	0	0	0	Merged — governing channel: Sideloaded rights; other component(s): Alternative app store access (final 0.0)	—	—
#7 Anti-steering	0	0	0	0	—	—	—
#8 Commercial independence	0.5	0.5	1	0.5	E=0.5: exclusivity ('choose-one-of-two') expressly banned; price-parity MFN less explicit; S=1: marketplace domain; Enf=0.5: guidelines + case-by-case AML enforcement (Alibaba 2021)	Marketplaces	SAMR Antitrust Guidelines for the Platform Economy Sector (February 2021) (choose-one-of-two ban); E-Commerce Law of the PRC (2019), Art. 35
#9 Platform access	0.5	0.5	0.5	0.5	E=0.5: fair-terms duties for merchants, not an access right; S=0.5: marketplaces only; Enf=0.5: administrative enforcement, discretionary	Marketplaces	E-Commerce Law of the PRC (2019), Art. 35; Anti-Unfair Competition Law of the PRC (2025 revision, eff. 15 October 2025) platform fair-rules duty
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	0.5	0.5	1	0.5	E=0.5: self-preferencing listed as potential abuse in guidelines; AUCL/Network Provisions ban technical disruption of rivals; S=1: platforms generally; Enf=0.5: administrative and case-by-case [Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.5)]	Platforms; Marketplaces, search, all recommendation services	SAMR Antitrust Guidelines for the Platform Economy Sector (February 2021); Anti-Unfair Competition Law of the PRC (2025 revision, eff. 15 October 2025) Art. 13; SAMR Interim Provisions on Anti-Unfair Competition on the Internet (eff. 1 September 2024) Art. 18 (eff. 2024-09-01) E-Commerce Law of the PRC (2019), Art. 40; CAC Provisions on the Administration of Algorithmic

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
							Recommendations for Internet Information Services (eff. 1 March 2022) Arts. 14-15 (eff. 2022-03-01); SAMR Interim Provisions on Anti-Unfair Competition on the Internet (eff. 1 September 2024) Art. 18 (eff. 2024-09-01)
#11 Service neutrality	0.5	0.5	1	0.5	E=0.5: tying via data/algorithms listed as abuse doctrine, not a per-se rule; S=1: platforms generally; Enf=0.5: case-by-case AML enforcement (Meituan, Alibaba) [Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.5)]	Platforms	Anti-Monopoly Law of the PRC (2022 amendment), Art. 22; SAMR Antitrust Guidelines for the Platform Economy Sector (February 2021) Anti-Monopoly Law of the PRC (2022 amendment); SAMR Antitrust Guidelines for the Platform Economy Sector (February 2021)
#12 Business user protection	0.5	0.5	1	0.5	E=0.5: platform duties over merchant data and a below-cost-coercion ban — adjacent, not a clean misuse prohibition; S=1: marketplace domain; Enf=0.5: administrative enforcement	Marketplaces	Anti-Unfair Competition Law of the PRC (2025 revision, eff. 15 October 2025) Arts. 13-14; E-Commerce Law
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	0	0	0	0	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.0)	—	—
#14 Network interoperability	0.5	0.5	0.5	0.5	E=0.5: administrative directive with weak formal status; S=0.5: link-blocking only — no protocol-level interoperability; Enf=0.5: window guidance, no private enforcement	Messaging	MIIT administrative directive on the blocking of external links (September 2021)
#15 Data accessibility	0	0	0	0	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)	—	—
#16 Cross-service data combination	0.5	0.5	1	0.5	PIPL Art. 6 confines processing to a clear and reasonable purpose directly related to it and to the minimum scope necessary, and Art. 23 requires separate consent before personal information is provided to another handler. Together these constrain cross-service combination without a competition-framed prohibition, so Existence is 0.5. Scope 1 as a rule of general application. Enforceability 0.5, consistent with the treatment of other Chinese cells: enforcement runs through administrative campaigns rather than a dedicated regulator applying penalties to this conduct.	All markets (PIPL)	Personal Information Protection Law (PIPL), Arts. 6 and 23
D1 average (Final Score)	0.40						
D2 average (Final Score)	0.25						
D3 average (Final Score)	0.50						
D4 average (Final Score)	0.25						

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
DE JURE INDEX (equal-weighted average of D1–D4)	0.35						

8. United Kingdom

DMCCA: obligations arise only as conduct requirements (CRs) imposed after SMS designation — designated-but-no-CR cells score 0 (machinery ready, nothing to enforce); CMA penalties up to 10% of global turnover. Three search CRs are in force at the observation date: the publisher conduct requirement, imposed 3 June 2026, and the fair-ranking and data-portability conduct requirements, both imposed 17 June 2026. The User Choice conduct requirement was consulted on 28 January 2026 and has not been imposed. Apple/Google mobile commitments took effect 1 April 2026. The assimilated P2B Regulation (2019/1150) provides an economy-wide business-user floor. DMCCA commitments are binding and CMA-enforceable, classified F2.

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0.5	0.5	E=0: choice screens sit in the User Choice CR, on which the CMA consulted on 28 January 2026 and which it has not imposed — the imposed search CRs are publisher (3 June 2026) and fair ranking and data portability (both 17 June 2026); S=0.5: addressees fixed by the SMS designation; Enf=0.5: machinery ready, nothing to enforce	—	DMCCA 2024 (Google search SMS designated Oct 2025; User Choice CR consulted 28 Jan 2026, not imposed)
#2 Default-setting restrictions	0	0	0.5	0.5	E=0: centralised default settings sit in the User Choice CR, on which the CMA consulted on 28 January 2026 and which it has not imposed — the imposed search CRs are publisher (3 June 2026) and fair ranking and data portability (both 17 June 2026); S=0.5: addressees fixed by the SMS designation; Enf=0.5: machinery ready, nothing to enforce	—	DMCCA 2024 (Google search SMS designated Oct 2025; User Choice CR consulted 28 Jan 2026, not imposed)
#3 Service removal	0	0	0.5	0.5	—	—	Digital Markets, Competition and Consumers Act 2024 (Apple/Google mobile SMS designated 22 Oct 2025; no uninstall CR imposed)
#4 Data portability	0.5	0.5	1	1	E=0.5: UK GDPR Art. 20 one-off portability only — continuous/API portability solely via the firm-specific Google Search Data Portability CR (imposed 17 Jun 2026); S=1: economy-wide; Enf=1: ICO + CMA	All markets (UK GDPR) + search (CR)	UK GDPR Art. 20; CMA Google Search Data Portability CR (imposed 17 Jun 2026)
#5 Choice integrity	0.5	0.5	1	1	E=0.5: unfair-commercial-practices regime (drip pricing, subscription traps, fake reviews) without naming dark patterns as a discrete category; S=1: all consumer-facing markets; Enf=1: CMA direct enforcement since Apr 2025	All consumer-facing markets	Digital Markets, Competition and Consumers Act 2024 Part 4 (unfair commercial practices; drip pricing; subscription contracts; fake reviews) in force Apr 2025
D2 — Rival Access & Distribution							
#6 Alternative distribution	0	0	0.5	0.5	Merged — governing channel: Sideloading rights; other component(s):	—	Digital Markets, Competition and Consumers

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
					Alternative app store access (final 0.0)		Act 2024 (mobile SMS designated; sideloading CR not yet imposed) Digital Markets, Competition and Consumers Act 2024 (mobile SMS designated; alt-store CR not yet imposed)
#7 Anti-steering	0	0	0.5	0.5	—	—	Digital Markets, Competition and Consumers Act 2024 (mobile SMS designated; steering CR not yet imposed)
#8 Commercial independence	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	— (no SMS CR; Competition Act 1998 ex-post only)
#9 Platform access	0	0	0.5	0.5	—	—	Digital Markets, Competition and Consumers Act 2024 (fair-dealing CR type available; none imposed)
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	0.5	1	0.5	1	E=1: Fair Ranking CR imposed 17 Jun 2026 (organic ranking incl. AI Overviews); Apple/Google mobile app-ranking commitments (1 Apr 2026) widen coverage; S=0.5: firm-specific; Enf=1: CMA penalties up to 10% global turnover	Search	DMCCA 2024 + Fair Ranking CR (17 Jun 2026); Apple/Google mobile app-ranking commitments (1 Apr 2026)
#11 Service neutrality	0	0	0.5	0.5	Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.0)	—	Digital Markets, Competition and Consumers Act 2024 (mobile SMS designated; no tying CR imposed) Digital Markets, Competition and Consumers Act 2024 (no CR imposed)
#12 Business user protection	0.5	1	0.5	1	E=1: Publisher CR (imposed 3 Jun 2026: gen-AI controls, engagement metrics, attribution) + Apple/Google mobile commitments (1 Apr 2026: app review, ranking, data use) + assimilated P2B Reg. 2019/1150; S=0.5: strongest channel firm-specific; Enf=1: CMA penalties for CR/commitment breach	Search; mobile; all intermediation (P2B)	CMA Google Search Publisher CR (3 Jun 2026); Apple/Google mobile commitments (1 Apr 2026); assimilated P2B Reg. 2019/1150
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	0.5	1	0.5	1	E=1: Apple mobile commitments (1 Apr 2026) — developer process to request interoperable access to iOS features, analogue of DMA Art. 6(7); DMCCA commitments binding with CMA fining powers (F2); S=0.5: Apple only; Enf=1: CMA penalties	Mobile OS (Apple)	DMCCA commitments re Apple mobile platform (interoperable access; 1 Apr 2026)
#14 Network interoperability	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	—
#15 Data accessibility	0	0	0.5	0.5	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)	—	Digital Markets, Competition and Consumers Act 2024 (no CR imposed) Digital Markets, Competition and Consumers Act 2024 (Google search designated; no query/click-data-sharing CR)

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#16 Cross-service data combination	0.5	0.5	1	1	The DMCCA framework empowers the CMA to impose conduct requirements on undertakings with Strategic Market Status, but of the four requirements consulted on for Google in February 2026 none addresses data combination; the regime is deliberately more incremental than the DMA on this point. What binds is UK GDPR purpose limitation and the lawful-basis requirement, which constrains cross-service combination without a competition-specific prohibition, so Existence is 0.5. Scope 1; Enforceability 1 through the Information Commissioner.	All markets (UK GDPR)	UK General Data Protection Regulation, Arts. 5 and 6; Digital Markets, Competition and Consumers Act 2024
D1 average (Final Score)	0.20						
D2 average (Final Score)	0.00						
D3 average (Final Score)	0.33						
D4 average (Final Score)	0.25						
DE JURE INDEX (equal-weighted average of D1–D4)	0.20						

9. Australia

Two instruments on point are in force: the Consumer Data Right (CCA Part IVD), enforced by the ACCC/OAIC, and Australian Privacy Principle 6 of the Privacy Act 1988, enforced by the OAIC. All other rows score 0: they are pipeline cells for the proposed Digital Competition Regime, except choice integrity (Unfair Trading Practices Bill 2026, pending) and commercial independence and network interoperability (no instrument on point).

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
D1 — User Mobility & Switching							
#1 Active choice	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#2 Default-setting restrictions	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#3 Service removal	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#4 Data portability	0.5	1	0.5	1	E=1: statutory CDR data-sharing right via mandated APIs; S=0.5: banking	Banking, energy	Consumer Data Right — Competition and

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
					and energy only — telecom designated but sharing pending; Enf=1: ACCC/OAIC enforcement	(telecom designated, sharing pending)	Consumer Act 2010 (Cth), Part IVD
#5 Choice integrity	0	0	0	0	—	—	Australian Consumer Law (general); Unfair Trading Practices Bill 2026 ('dark patterns' on the grey list)
D2 — Rival Access & Distribution							
#6 Alternative distribution	0	0	0	0	Merged — governing channel: Sideloaded rights; other component(s): Alternative app store access (final 0.0)	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#7 Anti-steering	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#8 Commercial independence	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	— (Competition and Consumer Act 2010 (Cth), s46 misuse of market power, ex-post only)
#9 Platform access	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
D3 — Platform Neutrality & Non-Discrimination							
#10 Ranking neutrality	0	0	0	0	Merged — governing channel: Self-preferencing prohibitions; other component(s): Ranking neutrality obligations (final 0.0)	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#11 Service neutrality	0	0	0	0	Merged — governing channel: Anti-tying provisions; other component(s): Anti-bundling provisions (final 0.0)	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#12 Business user protection	0	0	0	0	—	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
D4 — Data, Interoperability & Infrastructure Access							
#13 Technical interoperability	0	0	0	0	Merged — governing channel: API access requirements; other component(s): Hardware interoperability obligations (final 0.0)	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted)
#14 Network interoperability	0	0	0	0	E=0: no instrument on point — general competition/consumer law only	—	—
#15 Data accessibility	0	0	0	0	Merged — governing channel: Business-user data access rights; other component(s): Search query and click-data access obligations (final 0.0)	—	Proposed Digital Competition Regime (Australian Treasury, announced 2 December 2024; not enacted) —

Provision (mechanism)	Final	E	S	Enf	Rationale (deviations only)	Target market / field	Legal basis (full name)
#16 Cross-service data combination	0.5	0.5	1	1	The proposed Digital Competition Regime is not enacted and imposes nothing. APP 6 confines use and disclosure of personal information to the primary purpose of collection, subject to a reasonably-expected related-purpose exception, which constrains cross-service combination only weakly and without any competition framing, so Existence is 0.5. Scope 1; Enforceability 1 through the Office of the Australian Information Commissioner. This is the weakest instance of the 0.5 band and is flagged for review.	All markets (Privacy Act 1988)	Privacy Act 1988 (Cth), Australian Privacy Principle 6
<i>D1 average (Final Score)</i>	<i>0.10</i>						
<i>D2 average (Final Score)</i>	<i>0.00</i>						
<i>D3 average (Final Score)</i>	<i>0.00</i>						
<i>D4 average (Final Score)</i>	<i>0.13</i>						
DE JURE INDEX (equal-weighted average of D1–D4)	0.06						

V. Market–Jurisdiction Applications

This section applies the framework market by market. It reports the De Facto Contestability layer of the Index: the observed contestability of specific gatekeeper–market pairs, measured by the extent to which enacted obligations have translated into changes in market conditions in practice. Where the De Jure layer scores the law as written across nine jurisdictions, the De Facto layer scores conduct and outcomes for individual Core Platform Services (CPS), comparing a pre-DMA baseline (December 2023) with the latest observed state (July 2026); the scoring criteria and evidence base it applies are set out in Section III. For each market, the section first characterizes the market itself, then presents the full De Facto scoring table for the gatekeeper–market pair — the sub-indicator scores, the pre/post comparison, and the supporting evidence for every cell — restates the market-level De Jure Contestability score — both as it stands in June 2026 and against a December 2023 pre-DMA baseline, so that the legal layer can be read as a movement rather than a level — and finally combines the layers into the market’s Effective Contestability. Two EU markets are coded in full, across three gatekeeper–market pairs: online search (Google Search), and app distribution (Apple App Store and Google Play). The De Jure baseline is coded for Google Search and the Apple App Store, the two pairs for which a market-level De Jure catalogue is reported.

A. EU — Online Search

Market: online search. Assessed service: Google Search (Alphabet Inc.), designated as a Core Platform Service under the EU Digital Markets Act (DMA) on 6 September 2023.

1. Market scope and applicable mechanisms

The standardized market is online search, assessed through Google Search, a designated Core Platform Service under the DMA. Applicable mechanisms are identified separately for the two layers. At the De Jure layer, all sixteen mechanisms are scored: in the European Union the DMA and the Platform-to-Business Regulation impose a complete set of obligations on the designated service. At the De Facto layer, the set narrows. De Jure scoring credits an obligation wherever it binds Alphabet as a designated gatekeeper, including where its operative locus is app distribution, the operating system or messaging; De Facto scoring observes only conduct in the search service itself, so a mechanism operating in an adjacent market has no object here. Technical interoperability (D4.1) and cross-platform messaging connectivity (D4.2) are therefore Not Applicable, and the remaining fourteen indicators apply. Applicability is fixed before scoring and held constant across both periods (Section III.C). Which of the fourteen could then be scored on the available evidence is a separate question, reported with the scoring table in Section V.A.3.

2. Market-level De Jure Contestability

For the search market, the Existence, Scope, and Enforceability factors are carried over from the jurisdiction-level coding of Section IV.B unchanged; for each mechanism, the governing Digital Markets Act or supporting provision and the rationale as applied to Google Search. Where an obligation directly

governs the search service it is justified on that provision; where it binds Alphabet as a designated gatekeeper but operates in an adjacent market — app distribution, the operating system, or messaging — it is marked as such and scored 1 as part of the gatekeeper's full obligation set. The table reports two periods. The June 2026 column states the law as it stands; the December 2023 column scores the same sixteen mechanisms against the pre-existing acquis with the Digital Markets Act excluded, so that what the Act contributed can be separated from what the Union already required. That baseline is a counterfactual rather than a literal legal state: at December 2023 the Act was in force, applicable, and the service had been designated, but the Article 5, 6 and 7 obligations did not bind until 7 March 2024, so scoring the literal state would set Enforceability to zero on every Act-based cell and reach the same result by a longer route. Applicability is held constant across the two periods, as Section III.C requires.

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
#1 Active choice	1 / 0.5 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. Decision AT.40099 Google Android (18 July 2018) ends the revenue-share payments conditioned on pre-installing no competing general search service; Google's compliance measure is the EEA search choice screen, live since March 2020 and revised in September 2021. Existence is 1 — an imposed conduct requirement on point, addressed to Google and Alphabet by name. Scope is 0.5: the decision binds one firm and reaches Android devices only, leaving the Safari and desktop default positions untouched. Enforceability is 1, with periodic penalty payments and the General Court's confirmation in September 2022. Jun 2026. Art. 6(3) makes the gatekeeper prompt users, on first use of its online search engine, to pick their default search provider from a choice screen and change it freely — a self-executing, Commission-enforced duty that names online search engines. Directly search-facing, so each factor is 1.
#2 Default-setting restrictions	0.5 / 0.5 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. The same decision reaches the contractual conditions that secured Google's default position, not the user's ability to change a default once set. No instrument requires the search default to be changeable through simple operations, so Existence is 0.5 for adjacent coverage; Scope is 0.5 for one firm on one operating system; Enforceability is 1. Jun 2026. Art. 6(3) also requires the search default to be changeable through simple operations; it applies directly to the search service and is enforced ex ante, so each factor is 1.
#3 Service removal	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force gives end users a right to remove pre-installed applications. AT.40099 governs OEM licensing conditions, not end-user removal. Each factor is 0. Jun 2026. Art. 6(3)'s duty to allow un-installation of pre-installed applications binds Alphabet as a gatekeeper but operates in the operating-system / app-distribution layer, not on the search service; scored 1 within the gatekeeper's full obligation set.

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
#4 Data portability	1 / 1 / 1 → 1	1 / 1 / 1 → 1	0	Dec 2023. GDPR Art. 20 confers a portability right squarely on point, its content fixed ex ante, binding every controller and so covering the mechanism's full economic domain. Supervisory authorities hold penalty powers up to EUR 20 000 000 or 4% of worldwide turnover, with private enforcement alongside. Each factor is 1 — the only mechanism the Union already covered completely in this market. Art. 6(9) DMA later raised the standard to continuous, real-time access; it did not create the right. Jun 2026. Art. 6(9), with GDPR Art. 20, requires effective, continuous, real-time portability of the data an end user generates in the search service, including search history; directly applicable to Google Search users, so each factor is 1.
#5 Choice integrity	0.5 / 1 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. DSA Art. 25, which bars deceptive or manipulative interface design, is addressed to providers of online platforms; Google Search is a designated very large online search engine and not an online platform within Art. 3, so Art. 25 does not reach it. What does reach it is the Art. 34 and 35 duty to assess and mitigate systemic risk, whose listed measures include adapting online interfaces. Existence is 0.5: a risk-based mitigation duty is adjacent to, not the same as, a prohibition. Scope is 1 and Enforceability 1 — exclusive Commission supervision with fines up to 6% of worldwide turnover, binding from 25 August 2023. Jun 2026. DSA Art. 25 bars deceptive or manipulative interface design and DMA Art. 13 bars conduct that circumvents compliance; both reach the search choice screen and consent flows, so each factor is 1.
#6 Alternative distribution	0.5 / 0.5 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. The anti-fragmentation infringement in AT.40099 frees hardware manufacturers to ship devices based on Android forks, opening a rival distribution channel at the operating-system layer. Existence is 0.5: it removes a restriction on rival operating-system distribution rather than conferring a right to third-party stores or sideloading. Scope 0.5 for one named firm; Enforceability 1. Jun 2026. Art. 6(4)'s duty to allow third-party apps and app stores binds the gatekeeper but operates in app distribution / the operating system, not in search; scored 1 within the full obligation set.
#7 Anti-steering	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force permits business users to promote or communicate off-platform offers. The Commission decision imposing that obligation is dated 4 March 2024, post-dates the baseline, and addresses Apple rather than Alphabet. P2B Art. 10 concerns pricing conditions and is scored at #8. Each factor is 0. Jun 2026. Art. 5(4) lets business users promote off-platform offers, but its operative locus is app stores and intermediation rather than the search service; scored 1 within the gatekeeper's full obligation set.
#8 Commercial independence	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Art. 10 requires the grounds for a parity restriction to be disclosed and made publicly available; it does not prohibit the restriction, and Art. 10(2) expressly

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
				leaves any prohibition to other Union or national law. Existence is 0.5 for disclosure in place of prohibition. Scope is 1 — all providers of online intermediation services, which is the mechanism's domain. Enforceability is 0.5: Art. 15 delegates enforcement to Member States, with no Union regulator holding penalty powers. Jun 2026. Art. 5(3)'s wide-parity (MFN) ban operates on online intermediation services, not on search; scored 1 within the gatekeeper's full obligation set.
#9 Platform access	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. For online search engines P2B imposes transparency alone. Arts. 3, 4, 11 and 12 — terms and conditions, statement of reasons, internal complaint-handling, mediation — are addressed to providers of online intermediation services and do not reach search engines, which receive only the ranking and differentiated-treatment descriptions of Arts. 5(2) and 7(2). Existence is 0.5: no fair, reasonable and non-discriminatory access standard and no dispute mechanism. Scope 1; Enforceability 0.5. Jun 2026. Art. 6(12) expressly requires fair, reasonable and non-discriminatory access for business users to the gatekeeper's online search engines, with published conditions and a dispute mechanism the Commission assesses. Search-specific, so each factor is 1.
#10 Ranking neutrality	1 / 0.5 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. Scored on the strongest available channel. P2B Art. 5(2) and UCPD Art. 7(4a) supply ranking transparency only; Decision AT.39740 Google Search (Shopping) (27 June 2017) supplies an operative obligation to treat rival comparison shopping services no less favourably in general search results. Existence is 1. Scope is 0.5 twice over: the decision binds one firm and reaches one vertical, leaving flights, hotels, local and the remainder uncovered. Enforceability is 1, upheld by the General Court in November 2021. Jun 2026. Art. 6(5) directly bars ranking, indexing or crawling the gatekeeper's own services more favourably than third parties and requires transparent, fair, non-discriminatory ranking (reinforced by P2B transparency) — the core search-ranking duty, so each factor is 1.
#11 Service neutrality	1 / 0.5 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. The first two of the four infringements in AT.40099 are tying abuses — the Google Search app and Chrome tied to the Play Store — which is the leveraging conduct this mechanism targets. Existence is 1. Scope is 0.5: one firm and one ecosystem, and the decision does not reach forced registration for further core platform services, nor browser-engine, identification or payment mandates. Enforceability 1. Jun 2026. Art. 5(8) bars requiring users to register for further Core Platform Services in order to use search, and Art. 5(7) bars forced use of the gatekeeper's browser engine, identification or payment services; the merged mechanism reaches search, so each factor is 1.
#12 Business user protection	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Art. 7(3)(a) requires any differential in the provider's own access to business-user data to be described. Existence is 0.5: the duty is to disclose the

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
				differential, not to refrain from using the data in competition with business users. Scope 1; Enforceability 0.5. Jun 2026. Art. 6(2) bars using business users' non-public data in competition with them and expressly names click, search, view and voice data — directly search-facing, so each factor is 1.
#13 Technical interoperability	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force requires access to operating-system, hardware or software features. The Data Act's interoperability provisions became applicable on 12 September 2025 and post-date the baseline. Each factor is 0. Jun 2026. Art. 6(7)'s interoperability access to operating-system, hardware and software features binds the gatekeeper but operates in the OS / device layer, not in search; scored 1 within the full obligation set.
#14 Network interoperability	0 / 0.5 / 0.5 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. EECC Art. 61(2) empowers national regulatory authorities to impose interoperability obligations on number-independent interpersonal communications services to secure end-to-end connectivity. It is a power, not an obligation: the content is not fixed in the Directive and it had not been exercised against any provider at the baseline date, BEREC's interoperability workstream running under Art. 7 DMA instead. This is the F5 signature — Existence 0 because no obligation is imposed on point, Scope and Enforceability 0.5 because the machinery exists but is untested. The minimum is 0. Jun 2026. Art. 7's number-independent messaging interoperability binds the gatekeeper but operates in the messaging market, not in search; scored 1 within the full obligation set.
#15 Data accessibility	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. P2B Art. 9, which requires business users' technical and contractual access to data to be described, is addressed to providers of online intermediation services and does not reach online search engines. Nothing in force gives corporate website users access to the ranking, query, click or view data they generate on the engine. Each factor is 0. Jun 2026. Art. 6(11) requires fair, reasonable and non-discriminatory access to ranking, query, click and view data generated by end users on the gatekeeper's search engine (anonymised where personal) — expressly search-specific, so each factor is 1.
#16 Cross-service data combination	0.5 / 1 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. No competition instrument restricted cross-service data combination at the baseline. What did bind was GDPR Art. 6, as interpreted in Case C-252/21 Meta Platforms v Bundeskartellamt (4 July 2023), which held that a dominant platform combining data across its services generally cannot rely on contract or legitimate interest and requires consent, and that a competition authority may assess GDPR compliance in an abuse case. Existence is 0.5: a lawful-basis requirement constrains combination without prohibiting it, and consent obtained through a bundled flow remains available. Scope 1: general application to every controller. Enforceability 1: supervisory

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
				authorities with penalty powers up to 4% of worldwide turnover. Jun 2026. Art. 5(2) bars the gatekeeper from combining personal data from the designated service with personal data from its other core platform services or any other service it provides, from cross-using such data, and from signing users in to other services in order to combine data, unless specific consent has been given. Directly applicable to the designated service, self-executing and Commission-enforced, so each factor is 1.

The sixteen mechanism scores aggregate as elsewhere in the Index: within each dimension the mechanism scores are averaged, and the four dimension scores are then averaged with equal weight into the market-level index (Section IV.A). All four dimensions score 1.00 — User Mobility & Switching (D1), from #1–#5; Rival Access & Distribution (D2), from #6–#9; Platform Neutrality & Non-Discrimination (D3), from #10–#12; and Data, Interoperability & Infrastructure Access (D4), from #13–#16 — giving an equal-weighted market-level De Jure Contestability of 1.00.

Dimension	Dec 2023	Jun 2026	Δ
D1 — User Mobility & Switching	0.50	1.00	+0.50
D2 — Rival Access & Distribution	0.38	1.00	+0.62
D3 — Platform Neutrality & Non-Discrimination	0.50	1.00	+0.50
D4 — Data, Interoperability & Infrastructure Access	0.13	1.00	+0.87
Market-level De Jure Contestability	0.38	1.00	+0.62

At the December 2023 baseline the same four dimensions stood at 0.50, 0.38, 0.50 and 0.13, giving an equal-weighted market-level De Jure Contestability of 0.38 — a movement of +0.62. One mechanism scored 1 before the Act: data portability, on Article 20 of the General Data Protection Regulation, which the Act raised to a continuous and real-time standard rather than created. Ten scored 0.5, held there by the Platform-to-Business Regulation's transparency-only character, by the firm-specific reach of an Article 102 remedy, or by the general terms of the data-protection acquis. Five scored 0, and D4 stood at 0.13: nothing in force required interoperability, messaging connectivity, or access to the data generated on the search engine, and cross-service data combination was constrained only by the lawful-basis requirement of Article 6 of the General Data Protection Regulation, as read in Case C-252/21 Meta Platforms v Bundeskartellamt.

3. Market-level De Facto Contestability

Gatekeeper: Alphabet Inc. Core Platform Service: Google Search. Designated 6 September 2023.

Two indicators are Not Applicable: technical and messaging interoperability operate in the device and communications layers. Two are Applicable but Missing. Twelve are scored.

Code	Indicator	Pre	Post	Δ	Evidence and scoring rationale
D1 — User Mobility & Switching					
D1.1	User switching behavior	0	0.5	+0.5	Google was the pre-selected default on all Android devices under MADA and in Safari under a paid default agreement, with no prompt to choose. Under earlier arrangements, choice screen placement was auctioned, with providers bidding per selection. From 6 March 2024, choice screens run in the EEA at Android setup and on first run of Chrome. From 7 August 2024 the search screen shows the top eight eligible providers ranked on Play install data, in randomised order with no pre-selection, and participation is free. The score is held at 0.5 because users must scroll the full list before selecting; the screen is shown once per device and again only after factory reset; it is displayed only where the OEM pre-installed the Google Search app and placed the search box on the home screen; participation requires application and approval in windows opening twice a year; and ten eligibility criteria apply, including a 5,000-install minimum. Google commits to a quarterly selection report to providers, per-country rankings, a Chrome URL parameter on searches routed by the screen, and the Play Install Referrer API.
D1.2	Default switching behavior	0.5	0.5	0	Changing the default engine in Chrome or Android settings was available throughout the pre-period, with no prompt. Google states that after the choice screen users can change Chrome's default or move, remove or replace the home screen search box, and that the screen text says so. No warning or confirmation dialog is imposed in either period.
D1.3	Application replacement behavior	0	0	0	The Google Search app and widget were pre-installed under MADA and could be disabled

					but not uninstalled. Post-DMA most Google applications can be removed on EEA builds, but Search cannot. Play Services remains non-removable on non-rooted devices, where the available actions are disabling it, clearing its cache, or removing updates.
D1.4	Data portability usage	0.5	0.5	0	Google Takeout exports activity records, including Search activity, in HTML and JSON, and was available throughout both periods. Google also operates a Data Portability API that gives authorised third parties access to user data through OAuth-scoped resource groups. The launch date of the API, and whether it meets the Art. 6(9) standard of continuous and real-time access, were not established in this coding round.
D1.5	Autonomous user decision-making	0	0.5	+0.5	No interface for choosing a search engine existed pre-DMA. The post-DMA choice screen uses randomised ordering with no pre-selection and free participation. Friction remains in single presentation per device and in Google's control of the screen's timing, placement and design on its own platform. The mandatory scroll-through is friction on the user and was introduced so that all options are seen.
	D1 average (5 of 5 scored)	0.2	0.4	+0.2	
D2 — Rival Access & Distribution					
D2.1	Alternative distribution adoption	0.5	0.5	0	Rival engines were reachable on the open web throughout both periods. The default position on Android and in Safari was contractually held by Google pre-DMA. The choice screen opens that position, Google operates it on its own platform, and Google confirms that providers may still conclude pre-installation deals with OEMs.
D2.2	Off-platform transactions	1	1	0	Art. 5(4) covers business users acquired via any core platform service, so the indicator applies to merchants, advertisers and

					publishers acquired through Search. The Shopping ads Policy Center covers prohibited content, prohibited practices, restricted content and site requirements, and none of the four restricts communicating or promoting offers to users or concluding contracts off-platform. Shopping routes the user to the merchant's own landing page and checkout. Two linked documents were not read: the Merchant Center Terms of Service and the Google Ads policies. The Commission's July 2026 anti-steering decision concerns Google Play, not Search.
D2.3	Commercial pricing flexibility	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Shopping sits inside the market boundary, so Art. 5(3) reaches it. Merchant Center policy requires the feed price to match the landing page and hold through checkout, enforced with disapproval and account suspension, and requires the landing page to show the same product regardless of the user's device, browser, location, cookies, ad targeting choices or any other consideration. No parity clause was found in the policy pages. Scoring requires the Merchant Center Terms of Service, which were not obtained. Excluded from the denominator.
D2.4	Rival participation	0.5	0.5	0	Publishers appear in Google Search in both periods, so a working path exists. On 13 November 2025 the Commission opened proceedings under Art. 6(12) and Art. 6(5) to assess whether Google applies fair, reasonable and non-discriminatory conditions of access to publishers' websites. The Commission's monitoring found indications that Google, under its site reputation abuse policy, demotes news media and other publishers' websites when they include content from commercial partners. Google states the policy targets ranking manipulation. The opening of proceedings does not prejudice a finding.
	D2 average (3 of 4 scored)	0.67	0.67	0	

D3 — Platform Neutrality & Non-Discrimination					
D3.1	Relative visibility of rival services	0	0.5	+0.5	Pre-DMA, the Google Flights and Google Hotels units carried availability and pricing on the results page alongside the Shopping Unit, with no dedicated placement for comparison services. Post-DMA, dedicated comparison-site units were introduced and the Flights unit removed ahead of the March 2024 deadline. Mirai measured Google Hotels' share as a source of hotel reservations falling from 13.4% to 8.9% in the EU, a relative decline of 33%, with online travel agencies gaining visibility within Google, against a non-EU control where the Google ecosystem was flat at -0.29%. The score is held at 0.5 by the Commission's decision of 23 July 2026 in Case DMA.100193, which found that Google gives its own shopping, hotels, transport and sports services greater prominence than comparable third-party services, including top placement and enhanced visuals and filters, and imposed a fine of EUR 460 million with a 60-day compliance period. The decision covers nine Alphabet services and dates the breach from 7 March 2024 to adoption. The Commission notes that Google has proposed and started testing changes to how it presents its free shopping, hotels and flights services, which it describes as substantial progress towards compliance and will monitor. Self-preferencing in AI Overviews and AI Mode is subject to continued dialogue. Google recaptured 3.5 of the 4.5 points lost by Google Hotels through its own paid and organic surfaces and retained 65.7% of direct hotel bookings.
D3.2	Independent service adoption	0.5	0.5	0	Search is usable without a Google account in both periods, so registration for further core platform services is not required to use it. A Google account links twelve or more services for signed-in users. No change is recorded post-DMA. The score rests on observed

					behaviour rather than documentation of Art. 5(7) and 5(8) compliance.
D3.3	Protection of business-user information	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Art. 6(2) names click, search, view and voice data. No breach has been found and no independent verification mechanism exists. Absence of an adverse finding does not establish the market condition. Excluded from the denominator.
	D3 average (2 of 3 scored)	0.25	0.5	+0.25	
D4 — Data, Interoperability & Infrastructure Access					
D4.1	Technical interoperability	<i>n/a</i>	<i>n/a</i>		NOT APPLICABLE. Art. 6(7) operates in the operating-system and device layer. The Commission's specification proceedings opened on 27 January 2026 under Art. 6(7) concern Android features used by Google's AI services, not the search service.
D4.2	Cross-platform messaging connectivity	<i>n/a</i>	<i>n/a</i>		NOT APPLICABLE. Art. 7 operates in the number-independent interpersonal communications market.
D4.3	Rivals and business-user data utilization	0	0.5	+0.5	Google is the only gatekeeper subject to Art. 6(11), which requires sharing anonymised ranking, query, click and view data with rival search engines on FRAND terms. The Commission opened specification proceedings DMA.100209 on 27 January 2026, addressed preliminary findings and draft measures to Alphabet on 16 April 2026, and adopted a final decision on 16 July 2026. The Commission states that two years of talks showed a specification process was the fastest way to make the dataset effective. The final measures require Google to begin sharing anonymised search data with eligible rivals, including AI chatbots with search functions, from January 2027, and to finalise its pricing offer by January 2027. The binding measures take effect after the observation window.

D4.4	Extent of incumbent data accumulation and combination	0	0.5	+0.5	Art. 5(2) consent flows for linking Google services were deployed in March 2024. In Case PS12714, opened in July 2024 and closed on 4 November 2025, the Italian AGCM accepted and made binding Google's commitments without finding an infringement. Google committed to state that linked services share personal data with each other, to list the services concerned, to state that most features continue to work if services are not linked, to make explicit that the decision can be deferred up to three times and that consent can be limited to some services, and to contact users who had already made a choice. The commitments also state that services outside the listed core platform services are always linked. Implementation was due within 180 days. The AGCM acted under Italian consumer protection law and noted that enforcement of Art. 5(2) is the competence of the Commission. Consent rates are not published.
	D4 average (2 of 4 scored)	0	0.5	+0.5	
	DE FACTO INDEX (equal-weighted D1–D4)	0.28	0.52	+0.24	

Google's De Facto Contestability Index rises from 0.28 to 0.52 ($\Delta +0.24$). D2 is the highest-scoring dimension in both periods at 0.67. Article 5(4) applies to business users acquired through Search, and the Shopping ads policies impose no restriction on communicating or promoting offers or on concluding contracts off-platform, so merchants could route customers to their own checkout before the DMA and can do so now, though further analysis might be warranted by pulling the full merchant center terms and services. The largest gain is in D4, which moves from 0 to 0.50 on the Article 5(2) consent flows deployed in March 2024 and on the European Search Dataset Licensing Program, which Google has operated since 6 March 2024. Under Article 6(11) the Commission opened specification proceedings in January 2026 and adopted binding measures on 16 July 2026; these take effect from January 2027, after the observation window. D3.1, self-preferencing in results, moves from 0 to 0.5: Google removed the Flights unit and introduced dedicated comparison-site units ahead of the March 2024 deadline, and Mirai measured Google Hotels' share as a source of reservations falling from 13.4% to 8.9%, a 33% relative decline, with

online travel agencies gaining visibility within Google against a flat non-EU control. The indicator is held below 1 by the Commission's decision of 23 July 2026 in Case DMA.100193, which found breach of Article 6(5) across nine Alphabet services, dated from March 2024 and ongoing at adoption, and imposed a fine of EUR 460 million with a 60-day remedy order. D1 moves from 0.20 to 0.40 on the EEA choice screens and their randomised, non-pre-selected design, held down by application replacement, which scores 0 in both periods because the Google Search app remains non-removable. Two indicators are excluded as Applicable but Missing: commercial pricing flexibility and protection of business-user information.

4. Effective Contestability

Gatekeeper–market (EU)	De Jure	De Facto	Effective Contestability
Google — Search	1.00	0.52	0.76

For Google in the EU search market, the market-level De Jure score is 1.00: at the De Jure layer the full catalogue of sixteen mechanisms applies and each scores 1 for the designated Google Search service. The De Facto Contestability Index stands at 0.52 as of July 2026 (Section V.A.3), so Effective Contestability is 0.76. The distance from 1 is an implementation shortfall, which Section V.A.3 identifies: the choice screen and the consent regime have improved user mobility and data conditions, while the structural barrier of a web-scale index and the still-unremedied self-preferencing in ranking hold the realized side down. With the legal ceiling already reached, every future De Facto gain of 0.1 raises the Effective Contestability score by 0.05.

Read against the December 2023 baseline, the same cell is a movement rather than a level. The legal layer stood at 0.38 and the realized layer at 0.28, giving a baseline Effective Contestability of 0.33. The legal layer has since moved +0.62 and the realized layer +0.24, so Effective Contestability has risen +0.43. Almost the whole of the legal movement is attributable to the Act: before it, ten of the sixteen mechanisms rested on transparency duties, on firm-specific Article 102 remedies, or on the general data-protection acquis, five rested on nothing, and only data portability was already complete.

B. EU — iOS App Distribution

Market: iOS app distribution. Assessed service: Apple App Store (Apple Inc.), designated as a Core Platform Service under the EU Digital Markets Act (DMA) on 6 September 2023.

1. Market scope and applicable mechanisms

The standardized market is iOS app distribution, assessed through the Apple App Store; the App Store and iOS are designated Core Platform Services under the DMA. Applicable mechanisms are identified separately for the two layers. At the De Jure layer, all sixteen mechanisms are scored: the DMA imposes a complete set of obligations on the designated services, so the market-level De Jure coding restates the full jurisdiction-level catalogue of Section IV. At the De Facto layer, the market has two gatekeepers. Under

Section III.D each is coded separately on the same indicator set rather than averaged: Apple Inc., for the App Store on iOS, and Alphabet Inc., for Google Play on Android. Little is filtered out here — the DMA’s distribution, steering, payment, ranking and interoperability obligations all act on conduct observable in this market — so only cross-platform messaging connectivity (D4.2) is Not Applicable and the remaining fifteen indicators apply to both gatekeepers. Applicability is fixed before scoring and held constant across both periods and both gatekeepers (Section III.C); the evidence available to score each indicator differs between the two, and is reported in Section V.B.3.

2. Market-level De Jure Contestability

For the app-distribution market, the Existence, Scope, and Enforceability factors are carried over from the jurisdiction-level coding of Section IV.B unchanged; for each mechanism, the governing Digital Markets Act or supporting provision and the rationale as applied to the Apple App Store. The Act's obligations govern app distribution directly across almost the whole catalogue; only messaging interoperability operates in an adjacent market, and it is marked as such. The table reports two periods. The June 2026 column states the law as it stands; the December 2023 column scores the same sixteen mechanisms against the pre-existing acquis with the Digital Markets Act excluded, so that what the Act contributed can be separated from what the Union already required. That baseline is a counterfactual rather than a literal legal state: at December 2023 the Act was in force, applicable, and the service had been designated, but the Article 5, 6 and 7 obligations did not bind until 7 March 2024, so scoring the literal state would set Enforceability to zero on every Act-based cell and reach the same result by a longer route. Applicability is held constant across the two periods, as Section III.C requires.

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
#1 Active choice	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force required Apple to present a choice screen for any service, and iOS carried none. Each factor is 0. Jun 2026. Art. 6(3) requires the gatekeeper to let users change defaults and, on iOS, to present browser and search choice screens — a self-executing, Commission-enforced duty on the designated App Store and iOS, so each factor is 1.
#2 Default-setting restrictions	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. Browser and mail defaults became changeable from iOS 14 as a product decision, not in response to any obligation. No instrument required it. Each factor is 0. Jun 2026. Art. 6(3) requires iOS defaults to be changeable through simple operations; directly applicable to the designated operating system, so each factor is 1.
#3 Service removal	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force gives end users a right to remove pre-installed applications; Safari, the App Store, Messages, Photos and Camera could not be removed. Each factor is 0. Jun 2026. Art. 6(3) requires that users be able to un-install any pre-installed application on iOS — a core app-distribution / operating-system duty, so each factor is 1.

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
#4 Data portability	1 / 1 / 1 → 1	1 / 1 / 1 → 1	0	Dec 2023. GDPR Art. 20 confers a portability right on point, its content fixed ex ante, binding every controller. Supervisory authorities hold penalty powers up to EUR 20 000 000 or 4% of worldwide turnover, with private enforcement alongside. Each factor is 1. Jun 2026. Art. 6(9), with GDPR Art. 20, requires effective, continuous, real-time portability of end-user data generated in the Core Platform Service; applicable to the designated services, so each factor is 1.
#5 Choice integrity	1 / 1 / 1 → 1	1 / 1 / 1 → 1	0	Dec 2023. The App Store is an online platform within DSA Art. 3 and was designated a very large online platform on 25 April 2023, so the Art. 25 prohibition on deceptive or manipulative interface design bound it from 25 August 2023. Existence is 1 — an operative prohibition on point with content fixed ex ante. Scope is 1 — addressed to all providers of online platforms and covering the assessed service in full. Enforceability is 1 — exclusive Commission supervision with fines up to 6% of worldwide turnover. The divergence from the search market on this mechanism follows from the Digital Services Act's architecture, not from any difference in the two firms' conduct. Jun 2026. DSA Art. 25 bars deceptive or manipulative interface design and DMA Art. 13 bars conduct that circumvents compliance; both reach App Store consent and installation flows, so each factor is 1.
#6 Alternative distribution	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force required Apple to permit third-party app stores, sideloading or web distribution, all of which it prohibited. Only Articles 101 and 102 TFEU applied, and no decision had issued. Each factor is 0. Jun 2026. Art. 6(4) requires the gatekeeper to allow third-party apps and app stores to be installed and used on, and reached outside, iOS — the core app-distribution duty, so each factor is 1.
#7 Anti-steering	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. The decision finding Apple's anti-steering provisions abusive is dated 4 March 2024 and post-dates the baseline; at December 2023 the case stood at a revised statement of objections, which imposes no obligation. P2B Art. 10 concerns pricing conditions and is scored at #8. Each factor is 0. Jun 2026. Art. 5(4) lets developers communicate and promote offers to, and contract with, users acquired through the App Store, free of charge — directly app-store-facing, so each factor is 1.
#8 Commercial independence	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. The App Store is an online intermediation service, so P2B Art. 10 applies directly: the grounds for restricting developers from offering the same items elsewhere under different conditions must be disclosed and made publicly available. Existence is 0.5 for disclosure in place of prohibition; Scope 1; Enforceability 0.5, enforcement being delegated to Member States by Art. 15. Jun 2026. Art. 5(3) bars preventing developers from offering the same items elsewhere at different prices or conditions (wide-party / MFN ban); directly applicable to App Store distribution, so each factor is 1.

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
#9 Platform access	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B applies to the App Store in full — terms and conditions under Art. 3, a statement of reasons on restriction, suspension or termination under Art. 4, internal complaint-handling under Art. 11, mediation under Art. 12. Existence is 0.5: procedural fairness, notice and dispute machinery, but no fair, reasonable and non-discriminatory access standard and no non-discrimination requirement on access terms. Scope 1; Enforceability 0.5. Jun 2026. Art. 6(12) expressly requires fair, reasonable and non-discriminatory access for business users to the gatekeeper's software application stores, with published conditions and a dispute mechanism — it names app stores, so each factor is 1.
#10 Ranking neutrality	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Art. 5(1) requires the main ranking parameters to be set out in the terms and conditions and Art. 7(1) a description of any differentiated treatment of the provider's own goods and services; UCPD Art. 7(4a) adds the same disclosure toward consumers from 28 May 2022. Existence is 0.5: transparency about self-preferencing, with no duty to rank neutrally, and no decision addressed App Store ranking at the baseline. Scope 1; Enforceability 0.5. Jun 2026. Art. 6(5) bars self-preferencing the gatekeeper's own apps in App Store ranking and requires transparent, fair, non-discriminatory ranking; directly applicable, so each factor is 1.
#11 Service neutrality	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Art. 6 requires the terms and conditions to state whether and on what conditions a business user may offer its own ancillary goods and services, which reaches the in-app purchase mandate as a disclosure duty without prohibiting the tie. Nothing in force addressed the WebKit browser-engine requirement or Apple ID binding. Existence 0.5; Scope 1; Enforceability 0.5. Jun 2026. Art. 5(7) bars requiring developers or users to use the gatekeeper's in-app payment, browser engine or identification service, and Art. 5(8) bars forced registration to further Core Platform Services — directly app-store-facing, so each factor is 1.
#12 Business user protection	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Arts. 9(2)(a) and 7(3)(a) require the provider's own access to business-user data to be stated, with the categories and conditions. Existence is 0.5: disclosure of access, with no restriction on using that data in competition with developers. Scope 1; Enforceability 0.5. Jun 2026. Art. 6(2) bars using developers' non-public data (sales, specification and other App Store data) in competition with them — directly app-store-facing, so each factor is 1.
#13 Technical interoperability	0 / 0 / 0 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. No instrument in force required access to iOS hardware or software features; browser engines other than WebKit were prohibited and NFC was reserved to Apple Pay. The mobile-payments commitments were made binding on 11 July 2024 and post-date the baseline. Each factor is 0. Jun 2026. Art. 6(7) requires interoperability with, and access to, the same iOS hardware and software features

Provision (mechanism)	Dec 2023 E/S/Enf → Score	Jun 2026 E/S/Enf → Score	Δ	Governing provision and Rationale
				the gatekeeper uses (operationalised in the March 2025 iOS interoperability decisions); directly applicable, so each factor is 1.
#14 Network interoperability	0 / 0.5 / 0.5 → 0	1 / 1 / 1 → 1	+1.0	Dec 2023. EEC Art. 61(2) empowers national regulatory authorities to impose interoperability obligations on number-independent interpersonal communications services to secure end-to-end connectivity, but the content is not fixed in the Directive and the power had not been exercised at the baseline date. The F5 signature applies: Existence 0, Scope and Enforceability 0.5, minimum 0. Jun 2026. Art. 7's number-independent messaging interoperability binds Apple as a gatekeeper but operates in the messaging market, not in app distribution; scored 1 within the gatekeeper's full obligation set.
#15 Data accessibility	0.5 / 1 / 0.5 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. P2B Art. 9 requires a description of business users' technical and contractual access to the data they and their consumers provide or generate, including in aggregated form. Existence is 0.5: the Article confers no access right, so a provider may comply by disclosing that access is withheld. Scope 1; Enforceability 0.5. Where the search market scores 0 on this mechanism, app distribution scores 0.5, because Art. 9 is addressed to online intermediation services alone. Jun 2026. Art. 6(10) requires effective, high-quality, continuous, real-time access for business users to the data they generate through the Core Platform Service; applicable to App Store developers, so each factor is 1.
#16 Cross-service data combination	0.5 / 1 / 1 → 0.5	1 / 1 / 1 → 1	+0.5	Dec 2023. No competition instrument restricted cross-service data combination at the baseline. What did bind was GDPR Art. 6, as interpreted in Case C-252/21 Meta Platforms v Bundeskartellamt (4 July 2023), which held that a dominant platform combining data across its services generally cannot rely on contract or legitimate interest and requires consent, and that a competition authority may assess GDPR compliance in an abuse case. Existence is 0.5: a lawful-basis requirement constrains combination without prohibiting it, and consent obtained through a bundled flow remains available. Scope 1: general application to every controller. Enforceability 1: supervisory authorities with penalty powers up to 4% of worldwide turnover. Jun 2026. Art. 5(2) bars the gatekeeper from combining personal data from the designated service with personal data from its other core platform services or any other service it provides, from cross-using such data, and from signing users in to other services in order to combine data, unless specific consent has been given. Directly applicable to the designated service, self-executing and Commission-enforced, so each factor is 1.

The sixteen mechanism scores aggregate as elsewhere in the Index: within each dimension the mechanism scores are averaged, and the four dimension scores are then averaged with equal weight into the market-level index (Section IV.A). All four dimensions score 1.00 — User Mobility & Switching (D1), from #1–#5; Rival Access & Distribution (D2), from #6–#9; Platform Neutrality & Non-Discrimination (D3),

from #10–#12; and Data, Interoperability & Infrastructure Access (D4), from #13–#16 — giving an equal-weighted market-level De Jure Contestability of 1.00.

Dimension	Dec 2023	Jun 2026	Δ
D1 — User Mobility & Switching	0.40	1.00	+0.60
D2 — Rival Access & Distribution	0.25	1.00	+0.75
D3 — Platform Neutrality & Non-Discrimination	0.50	1.00	+0.50
D4 — Data, Interoperability & Infrastructure Access	0.25	1.00	+0.75
Market-level De Jure Contestability	0.35	1.00	+0.65

At the December 2023 baseline the four dimensions stood at 0.40, 0.25, 0.50 and 0.25, giving an equal-weighted market-level De Jure Contestability of 0.35 — a movement of +0.65. No Article 102 decision bound Apple in the Union at that date: the music-streaming decision is dated 4 March 2024 and the mobile-payments commitments 11 July 2024. Apple's baseline therefore rests entirely on instruments of general application, but on more of them than search does, because the App Store is both an online intermediation service under the Platform-to-Business Regulation and an online platform under the Digital Services Act, and an online search engine is neither. Two mechanisms scored 1, seven scored 0.5 — six of them capped by Article 15, which delegates enforcement of that Regulation to Member States, the seventh resting on the lawful-basis requirement of the data-protection acquis — and seven scored 0, precisely the channels the Act was written to open.

3. Market-level De Facto Contestability

EU — iOS App Distribution

Gatekeeper: Apple Inc. Core platform service: iOS operating system and App Store (online intermediation). Designated 6 September 2023. Baseline December 2023; post-period July 2026.

Code	Indicator	Pre	Post	Δ	Evidence and scoring rationale
D1 — User Mobility & Switching					
D1.1	User switching behavior	0	0.5	+0.5	No choice screen existed on iOS before the DMA. A browser choice screen appeared on first launch of Safari after updating to iOS 17.4 in March 2024. In iOS 18, following dialogue with the Commission, Apple redesigned the screen and showed it again to every EU user with Safari as default, on each of their devices,

					and again on migration to a new device with Safari as default. The Commission closed its Art. 6(3) investigation without a fine on 23 April 2025. No choice screen is presented for app distribution.
D1.2	Default switching behavior	0	0.5	+0.5	Before the DMA only browser and mail defaults were changeable, from iOS 14. By the observation date EU users could set defaults for browser, mail, app marketplace, contactless payments, phone calls, messaging, password managers, keyboards and call spam filters, with navigation and translation added in spring 2025, managed from a Default Apps section in Settings. Changing any default other than the browser requires navigating Settings with no prompt, which is the basis on which the equivalent Android and Search cells score 0.5.
D1.3	Application replacement behavior	0	1	+1	Safari, App Store, Messages, Photos and Camera could not be removed before the DMA. With iOS 18.2, EU users can fully delete them; almost all apps can be deleted except Phone and Settings, and deleted apps can be reinstalled from an App Installation section in Settings. Phone and Settings are plausibly within the Art. 6(3) carve-out for apps essential to the operating system. This is the only score of 1 in any of the three observations.
D1.4	Data portability usage	0.5	0.5	0	Batch export through privacy.apple.com existed throughout, with no ingestion path into a rival ecosystem. A joint Apple and Google cross-OS transfer tool was announced in December 2025, alongside AppMigrationKit and eSIM transfer. Export remains batch, so the Art. 6(9) continuous and real-time standard is not met.
D1.5	Autonomous user decision-making	0	0.5	+0.5	No alternative distribution channel existed before the DMA. The marketplace installation flow introduced in March 2024 ran to 15 steps, with a warning screen and a dead-end in Settings. In preliminary findings of 23 April 2025 under Art. 6(4), the Commission found the

					process overly burdensome and confusing. With iOS 18.6 in July 2025 Apple reduced the flow to six steps and removed the warning screen and the Settings dead-end. Epic Games reports that drop-off during installation of its store fell from about 65% to about 25%. Apple has announced a further update to the installation experience for autumn 2026, outside the observation window.
	D1 average (5 of 5 scored)	0.1	0.6	+0.5	
D2 — Rival Access & Distribution					
D2.1	Alternative distribution adoption	0	0.5	+0.5	Third-party app stores were prohibited on iOS before the DMA. By the observation date alternative marketplaces including AltStore PAL and the Epic Games Store operated in the EU, and web distribution was permitted. Setapp Mobile closed on 16 February 2026, with MacPaw citing still-evolving and complex business terms. To operate a marketplace, a developer had to provide a EUR 1,000,000 stand-by letter of credit from an A-rated institution, or have two years' good standing in the Developer Program and an app with more than one million first annual EU installs in the prior year. A marketplace app paid the Core Technology Fee of EUR 0.50 from its first annual install; AltStore PAL charged users EUR 1.50 a year to cover it until an Epic Games MegaGrant in August 2024. Notarisation by Apple is required on every channel. In its preliminary findings of 23 April 2025 under Art. 6(4), the Commission cited the fee, the eligibility requirements and the installation flow. Apple announced new EU terms on 18 August 2026, effective 1 October 2026, which widen eligibility and replace the Core Technology Fee; these fall outside the observation window.
D2.2	Off-platform transactions	0	0.5	+0.5	Link-outs and communication of external offers were prohibited before the DMA. On 23 April

					2025 the Commission fined Apple EUR 500 million under Art. 5(4) and ordered it to remove technical and commercial restrictions on steering. Apple's June 2025 terms permit communication and promotion of offers subject to a 2% initial acquisition fee, a store services fee of 5% or 13% depending on tier, and, for apps under the external purchase link addendum, a 5% Core Technology Commission. Apple's planned move to a single business model by January 2026 did not take effect during the observation window. Its August 2026 announcement replacing these fees takes effect on 1 October 2026.
D2.3	Commercial pricing flexibility	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Scoring requires the operative text of the Apple Developer Program Licence Agreement and App Store Review Guidelines on pricing and terms parity. Enforcement has proceeded under Art. 5(4) rather than Art. 5(3), which does not establish the contract terms. Excluded from the denominator.
D2.4	Rival participation	0	0.5	+0.5	App Review operated on discretionary criteria with no published access terms and no dispute mechanism. Apple now publishes Art. 6(12) access conditions and operates a dispute process. Notarisation is required on every distribution channel, and opting out of Tier 2 services forfeits discovery, search placement and featuring.
	D2 average (3 of 4 scored)	0	0.5	+0.5	
D3 — Platform Neutrality & Non-Discrimination					
D3.1	Relative visibility of rival services	0	0.5	+0.5	Apple applications were pre-installed and editorially promoted with no published ranking criteria. Ranking parameters are now published under Art. 6(5). No independent audit of App Store search neutrality exists, and Commission enforcement against Apple has addressed

					steering rather than ranking.
D3.2	Independent service adoption	0	0.5	+0.5	An Apple ID bound the App Store to iCloud, Apple Pay, Messages and FaceTime, and Apple in-app purchase was mandatory for digital goods. Alternative payment service providers are now permitted and core applications are deletable. The account structure is unchanged and iMessage remains exclusive to Apple hardware.
D3.3	Protection of business-user information	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Art. 6(2) applies and no breach has been found. No independent verification mechanism exists. Excluded from the denominator.
	D3 average (2 of 3 scored)	0	0.5	+0.5	
D4 — Data, Interoperability & Infrastructure Access					
D4.1	Technical interoperability	0	0.5	+0.5	Before the DMA, NFC was reserved to Apple Pay and third-party devices had no access to iOS connectivity features such as notification forwarding and proximity pairing. Apple opened NFC to third-party payment apps in the EU from iOS 17.4. The Commission's specification decisions of 19 March 2025 (DMA.100203 and DMA.100204) require notification forwarding, proximity pairing and other connectivity features, with most measures due by 1 June 2026 and audio switching to non-connected devices by 1 June 2027. In iOS 26.3 Apple forwards notifications to only one third-party device at a time, and enabling a third-party device turns off notifications to an Apple Watch.
D4.2	Cross-platform messaging connectivity	<i>n/a</i>	<i>n/a</i>		NOT APPLICABLE. Art. 7 operates in the number-independent interpersonal communications market.
D4.3	Rivals and business-user data utilization	0.5	0.5	0	App Store Connect provided download, revenue and crash reporting throughout the pre-period, with ranking signals withheld. Browser developers now receive data on the

					performance of the browser choice screen. Ranking signals remain withheld.
D4.4	Extent of incumbent data accumulation and combination	0.5	0.5	0	App Tracking Transparency required opt-in for cross-application tracking from 2021, and Apple applied on-device processing to parts of its data handling. Art. 5(2) applies from March 2024. Apple's advertising business is small relative to other gatekeepers, so the marginal restriction is limited.
	D4 average (3 of 4 scored)	0.33	0.5	+0.17	
	DE FACTO INDEX (equal-weighted D1–D4)	0.11	0.53	+0.42	

Google Play App Distribution

Gatekeeper: Alphabet Inc. Core Platform Services: Android and Google Play. Designated 6 September 2023.

One indicator is Not Applicable and five are Applicable but Missing. Ten are scored. The market definition and indicator set are identical to those used in Section V.B.

Code	Indicator	Pre	Post	Δ	Evidence and scoring rationale
D1 — User Mobility & Switching					
D1.1	User switching behavior	0	0.5	+0.5	Google Search and Chrome were pre-selected on Android devices under MADA, with no choice screen. Search and browser choice screens were deployed in the EEA from March 2024 in randomised order with no pre-selection, shown once per device. No choice screen exists for app distribution.
D1.2	Default switching behavior	0.5	0.5	0	Android's default apps setting has covered browser, phone, SMS, digital assistant, launcher, caller ID and wallet since before the DMA, and the category list is documented in Google's Android Management API reference.

					Changing a default requires navigating settings with no prompt. The EEA choice screens add prompting rather than new categories.
D1.3	Application replacement behavior	0	0	0	Pre-installed GMS applications under MADA could be disabled but not uninstalled. Google Play Services remains a protected system component: on non-rooted devices it cannot be uninstalled, and the available actions are disabling it, which breaks most Google applications, clearing its cache, or removing updates to revert to the factory version. Whether this falls within the Art. 6(3) carve-out for applications essential to the operating system is not settled by any published Commission assessment.
D1.4	Data portability usage	0.5	0.5	0	Google Takeout exports activity records in HTML and JSON and was available before the DMA. Google also operates a Data Portability API giving authorised third parties access to user data through OAuth-scoped resource groups. Whether this meets the Art. 6(9) standard of continuous and real-time access was not established.
D1.5	Autonomous user decision-making	0.5	0.5	0	Sideloads were available throughout behind a per-source install permission introduced in Android 8.0 in 2017, replacing the earlier global Unknown Sources setting. Epic Games describes Google's installation flow for its store as 12 steps with a warning that software from reputable competitors may be harmful, and reports that it defeats more than half of installation attempts; Epic is an interested party. The Registered App Stores programme announced on 4 March 2026 is planned to launch with a major Android release by the end of 2026 and was not operative during the observation window.
	D1 average (5 of 5 scored)	0.3	0.4	+0.1	
D2 — Rival Access & Distribution					

D2.1	Alternative distribution adoption	0.5	0.5	0	Android permitted sideloading and alternative stores throughout the pre-period. In its decision of 23 July 2026 the Commission found that Google prevents app developers from freely communicating and promoting offers and concluding contracts with users in distribution channels of their choice, including third-party app stores, and ordered Google to allow this both technically and contractually. The Registered App Stores programme was not operative during the observation window.
D2.2	Off-platform transactions	0	0.5	+0.5	Play Billing was mandatory for digital goods and direct links to external payment were prohibited before the DMA. From 30 June 2026 Google separated its service fee from its billing fee in the EEA. The service fee starts at 10% on the first USD 1 million of annual earnings and applies whether the developer uses Play billing, alternative billing or external web links; a 5% billing fee applies only when Play billing is used. On 23 July 2026 the Commission fined Google EUR 430 million under Art. 5(4), finding that Google prevents developers from freely communicating offers and concluding contracts in channels of their choice, and that the level of steering-related fees and the length of the charging period went beyond what is compliant. The Commission notes that Google's rolled-out changes to its steering terms constitute good progress towards compliance. A 60-day compliance period applies.
D2.3	Commercial pricing flexibility	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Scoring requires the operative text of the Play Developer Distribution Agreement on pricing and terms parity. Excluded from the denominator.
D2.4	Rival participation	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Scoring requires the pre-DMA Play access and appeals terms and the published Art. 6(12) EEA terms. Not

					obtained. Excluded from the denominator.
	D2 average (2 of 4 scored)	0.25	0.5	+0.25	
D3 — Platform Neutrality & Non-Discrimination					
D3.1	Relative visibility of rival services	0	0.5	+0.5	Google applications were pre-installed and promoted within Play with no published ranking criteria. Ranking parameters are now published under Art. 6(5). The Commission's July 2026 self-preferencing decision addressed Google Search, not Play ranking. No Play-specific finding and no independent audit exist.
D3.2	Independent service adoption	0	0.5	+0.5	Google Mobile Services were licensed as a bundle with pre-installation conditions under MADA, and Play Billing was mandatory for digital goods. On 2 July 2026 the Court of Justice dismissed Google's appeal in Case C-738/22 P, confirming that the pre-installation and licensing conditions were an abuse of dominance and upholding a fine of EUR 4.125 billion; this establishes the pre-period conduct. From 30 June 2026 the billing fee applies only when Play billing is used, while the service fee applies to all payment routes.
D3.3	Protection of business-user information	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Art. 6(2) applies and no breach has been found. No independent verification mechanism exists. Excluded from the denominator.
	D3 average (2 of 3 scored)	0	0.5	+0.5	
D4 — Data, Interoperability & Infrastructure Access					
D4.1	Technical interoperability	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Scoring requires documentation of Android's pre-period position on third-party browser engines, NFC access and public API scope. Not obtained. Excluded from the denominator.

D4.2	Cross-platform messaging connectivity	<i>n/a</i>	<i>n/a</i>		NOT APPLICABLE. Art. 7 operates in the number-independent interpersonal communications market.
D4.3	Rivals and business-user data utilization	<i>missing</i>	<i>missing</i>		APPLICABLE BUT MISSING. Scoring requires a documented inventory of Play Console reporting in the pre-period and of the Art. 6(10) extensions. Not obtained. Excluded from the denominator.
D4.4	Extent of incumbent data accumulation and combination	0	0.5	+0.5	Art. 5(2) consent flows for linking Google services were deployed in March 2024. In Case PS12714, closed on 4 November 2025, the AGCM accepted and made binding Google's commitments to revise the consent request, without finding an infringement. The commitments state that services outside the listed core platform services are always linked. Implementation was due within 180 days. Consent rates are not published.
	D4 average (1 of 4 scored)	0	0.5	+0.5	
	DE FACTO INDEX (equal-weighted D1–D4)	0.14	0.48	+0.34	

Apple's De Facto Contestability Index rises from 0.11 to 0.53 ($\Delta +0.42$) and Google Play's from 0.14 to 0.48 ($\Delta +0.34$). Apple's pre-period score is the lowest of the three observations because it prohibited alternative distribution, link-outs and removal of pre-installed applications outright, leaving D2 and D3 at 0. Its gain is concentrated in D1, which reaches 0.60, the largest dimension movement in any observation ($+0.50$), carried by the only indicator to reach 1 through post-DMA change: from iOS 18.2, EU users can fully delete Safari and the App Store. Google Play starts higher on D1 and D2 because Android permitted sideloading behind a per-source permission from Android 8.0 and exposed centralised default-app management before the DMA. Its movement is concentrated in the two channels Android had closed: the restructure of 30 June 2026 separated the service fee, which applies to every payment route, from a billing fee that applies only when Play billing is used, and Article 5(2) consent flows for linking Google services were deployed in March 2024. Alternative distribution does not rise: Android permitted sideloading and alternative stores before the DMA, and the Registered App Stores programme announced on 4 March 2026 was not operative during the observation window. In its decision of 23 July 2026, which fined Google

EUR 430 million under Article 5(4), the Commission found that Google prevents developers from freely communicating and promoting offers and concluding contracts in distribution channels of their choice, including third-party app stores. The two indices do not rest on equal evidence: five of Google Play's sixteen indicators are excluded as Applicable but Missing against two for Apple, and four of the five reflect documentation not obtained in this coding round rather than any feature of the market. Google Play's index therefore rests on a thinner evidence base than Apple's, not on a narrower set of applicable mechanisms.

4. Effective Contestability

Gatekeeper–market (EU)	De Jure	De Facto	Effective Contestability
Apple — iOS App Distribution	1.00	0.53	0.77
Google — Android App Distribution	1.00	0.48	0.74

For Apple in the EU iOS app-distribution market, all sixteen mechanisms operate and each scores 1 for the designated App Store, giving a market-level De Jure score of 1.00. The De Facto Contestability Index stands at 0.53 (Section V.B.3), so Effective Contestability is 0.77 — the higher of the two cells, because the DMA's distribution obligations act directly on platform-imposed prohibitions whose removal is observable: third-party stores exist and core apps can be uninstalled, even though adoption frictions keep the realized side at half strength.

Against the December 2023 baseline, Apple's legal layer stood at 0.35 and its realized layer at 0.11, giving a baseline Effective Contestability of 0.23. The legal layer has moved +0.65 and the realized layer +0.42, so Effective Contestability has risen +0.54, the largest movement recorded in the Index. The two markets' legal layers moved almost identically, +0.62 in search and +0.65 in app distribution, although their pre-existing regimes were differently composed — Alphabet's baseline rested substantially on firm-specific Article 102 remedies, Apple's entirely on instruments of general application. The realized layers did not move alike, and the implementation shortfall is roughly three times larger in search. The baselines explain why: Apple's starting position was one of outright prohibition, and prohibitions are cheap to remove and easy to observe being removed, whereas the search conditions the Act targets were already partially regulated in 2023 by the Shopping and Android decisions and have moved comparatively little since. Google Play has a De Facto baseline but no De Jure baseline, so its cell is reported as a level only.